Robot Sensing

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Reference Books:

- Probabilistic Robotics. S. Thrun, W. Burgard, D. Fox. MIT Press. 2001
- Simultaneous Localization and Mapping for Mobile Robots: Introduction and Methods. Juan-Antonio Fernández-Madrigal and José Luis Blanco Claraco. IGI-Global. 2013.

Contents

- Kinds of sensors for robotics
  - Classification
  - GPS
  - Sonars
  - Lidars
  - Cameras
- Probabilistic Sensor Models
  - Landmark sensor models
  - Scan-based sensor models

Classification of Sensors

We will focus on these

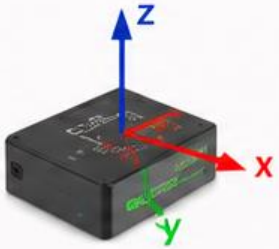
Proprioceptive sensors

- Measure the robot's **internal state** or **motion**:
  - battery, wheel position, acceleration, inclination, ...

Examples



Shaft encoder
(wheel position)



IMU
(accelerometer + gyroscope)



Potentiometer
(joint angle)



Inclinometer
(inclination)



Battery sensor
(state of charge)

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Exteroceptive sensors

- Measure properties of the **external environment**:
  - distance and/or angle to objects, light intensity reflected by objects, ...

Examples



Camera
(2D/3D vision)



Laser scanner (LiDAR)
(distance)



RGB-D camera
(depth + color)



Sonar
(distance)



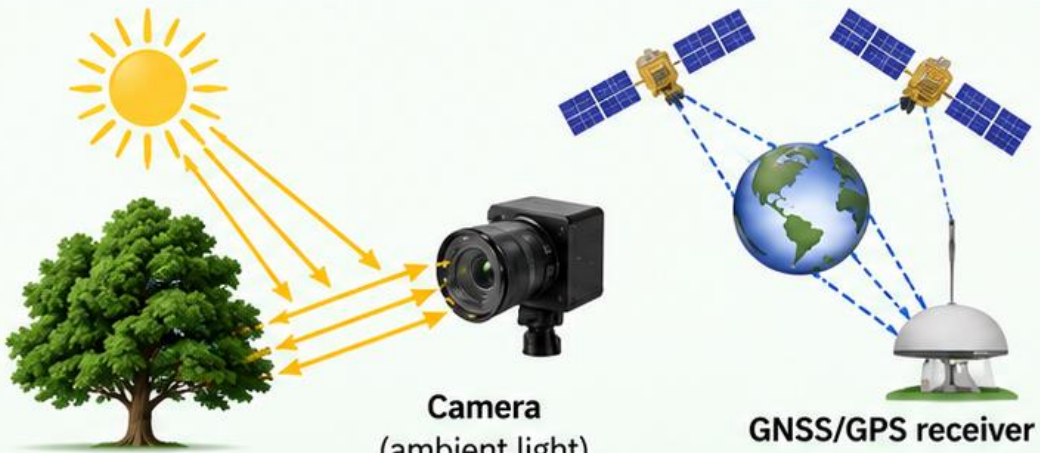
Infrared
(distance)

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Exteroceptive Sensors

Passive sensors

- Use **energy** that comes **from the environment**.
- **Examples**: cameras, GNSS/GPS receiver, ...



Examples



Camera
(2D/3D vision)

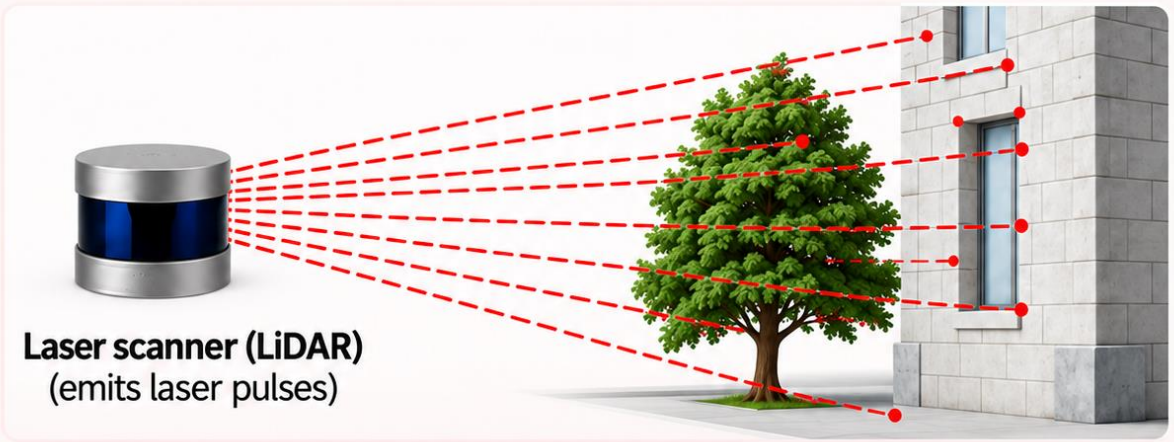


GNSS/GPS receiver
(satellite signals)

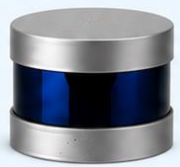
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Active sensors

- Emit their **own energy** (e.g. light, radio, sound).
- Better performance in changing-light conditions, but **more power consumption.**



Examples



Laser scanner (LiDAR)
(distance)



Sonar
(distance)



RGB-D camera
(depth + color)

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Landmarks and Beacons

Landmark

A **recognizable feature or object** of the environment whose observation can be associated with an element of a map.



Object (building)



Natural feature (mountain)



Distinctive 3D key points

Advantages

- **No installation required**
  Reuse existing elements of the environment.
- **Flexible and adaptable**
  to changing environments.

Beacon (artificial landmark)

An **intentionally installed or designated** landmark to facilitate detection, identification and localization.



GPS satellite
(known position)



Visual pattern
(QR, datamatrix, ...)



Reflector



Radio/WiFi antenna

Advantages

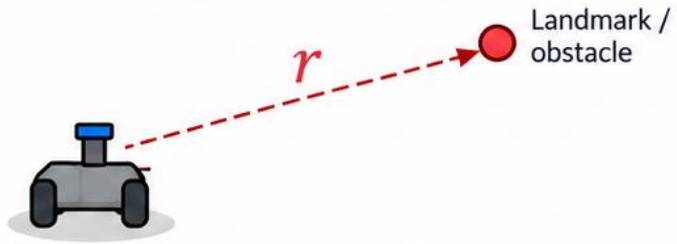
- **Provide absolute positioning**
  (with respect to known beacon coordinates).
- **Enable simple and efficient algorithms.**

Geometric measurements in robotics

Range

distance to an observed object/landmark

$$z = r$$

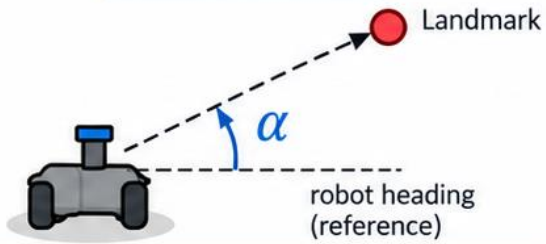


Examples: sonar, LiDAR, GPS pseudorange

Bearing

direction to a landmark

$$z = \alpha$$

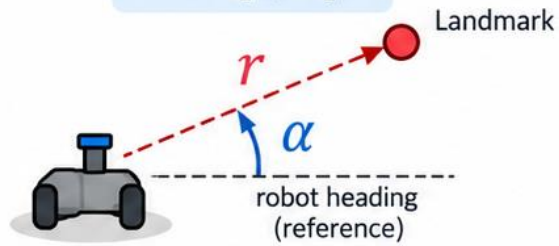


Example: monocular camera

Range + bearing

both quantities

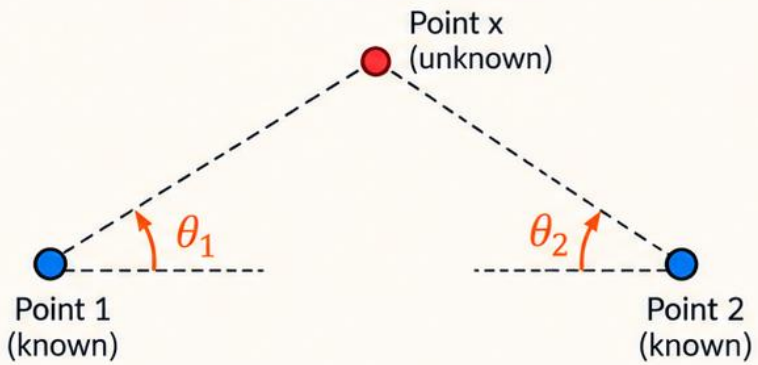
$$z = [r, \alpha]^T$$



Examples: 2D LiDAR observation, stereo/RGB-D landmark observation

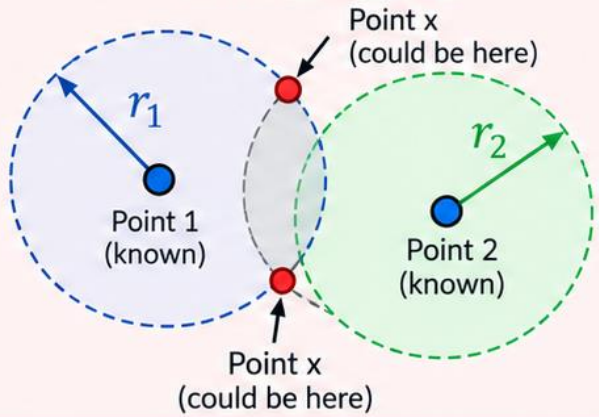
Triangulation

Determine the location of a point by measuring angles to it from known beacons.



Trilateration

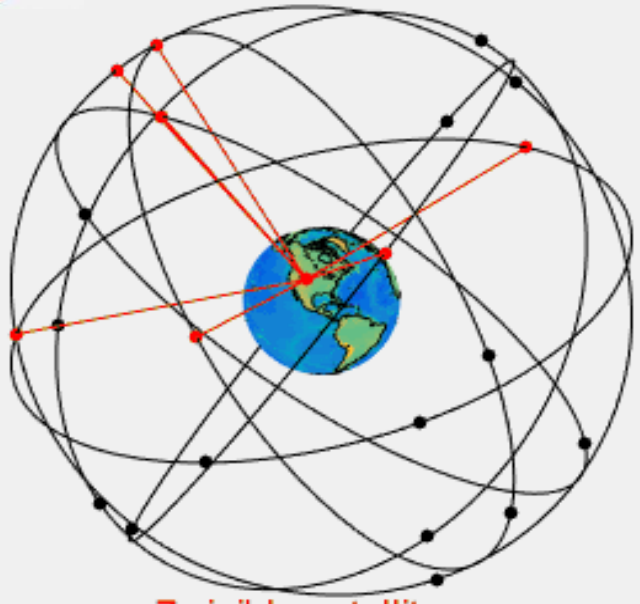
Determine the location of a point by measuring distances to known beacons.



Global Positioning System (GPS)

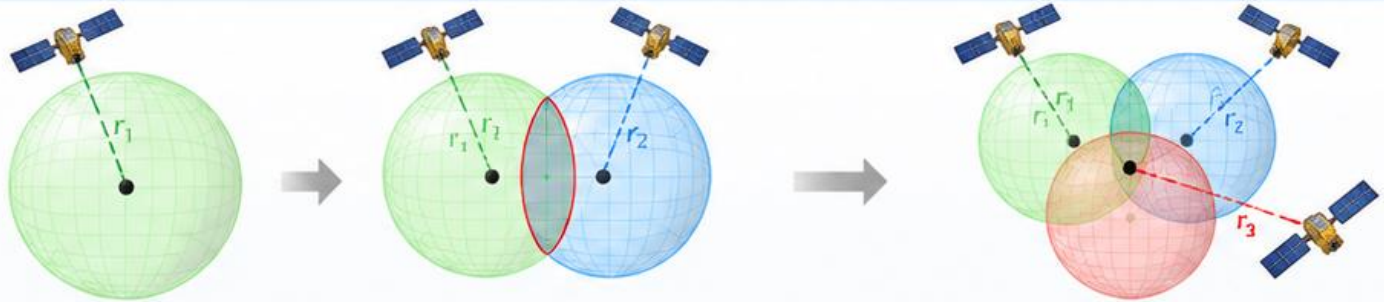
Widely used for **outdoor robot positioning**

1. Satellites act as **moving artificial beacons**
   - Transmit time-stamped signals and have known positions.
2. Satellites orbit the Earth at ≈ **20,200 km** altitude
   - Orbital period: about **12 hours** (six orbital planes).
3. Positioning is based on **trilateration**:
   determine the position from range measurements (**pseudoranges**) to multiple satellites.
4. At least **four satellites** are needed
   to estimate the 3D position and the receiver's clock offset.



Trilateration (range-based positioning)

The receiver position is at the intersection of the spheres defined by the measured distances to the satellites.



1 satellite
→ sphere

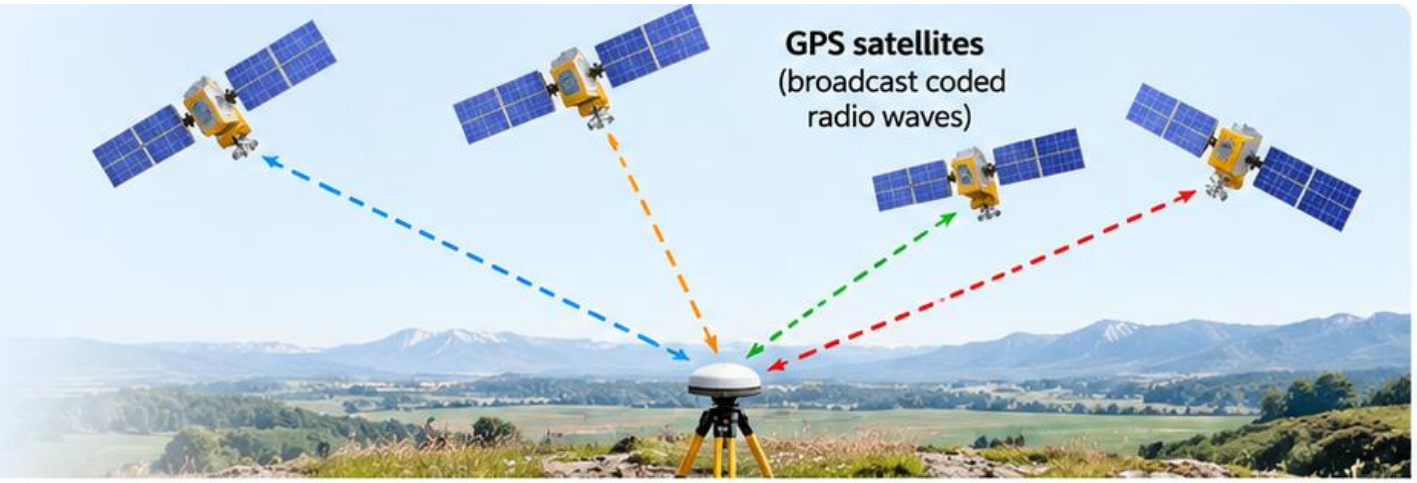
2 satellites
→ circle intersection

3 satellites
→ point intersection
(ideal case)

Global Positioning System (GPS)

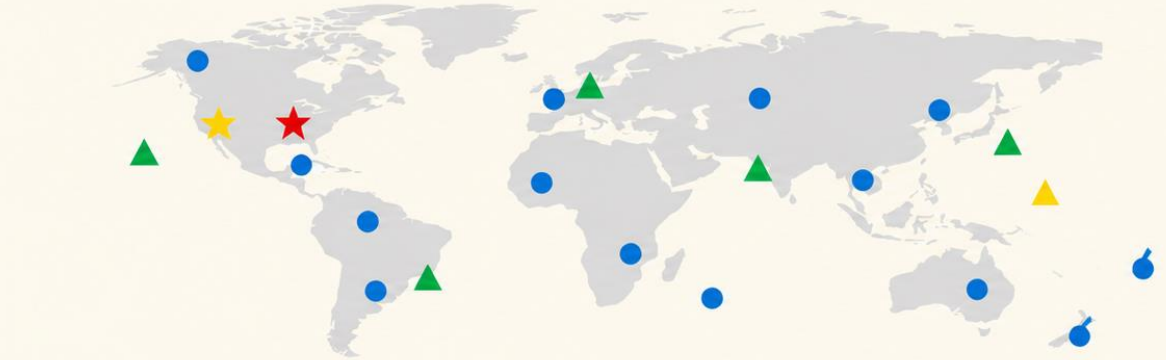
Satellite signal broadcast

- Each satellite broadcasts **coded radio waves** (several frequencies) containing:
  - Identity and location of the satellite
  - Data and time when the signal was sent (precise atomic clock)



GPS control segment

- Global network of **ground facilities** that monitor the GPS satellites and send corrections and commands.



1 Monitoring stations

Track satellites and monitor signals.



2 Master Control Station (MCS)

Compute corrections and generate commands.



3 Upload to satellites

Send corrections and commands to the satellites (via ground antennas).

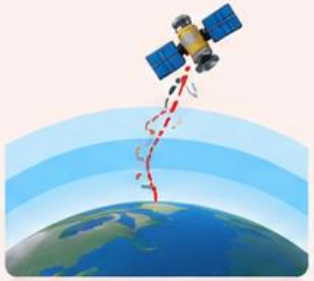


Global Positioning System (GPS). Errors

Main sources of error



Time synchronization



Atmospheric effects
ionosphere / troposphere

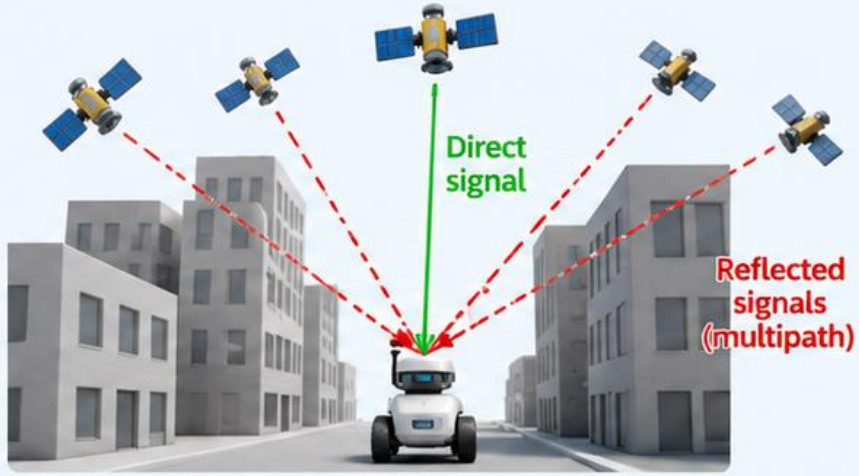


Multipath
reflections from buildings, terrain, ...



Limited satellite visibility
(e.g. trees, canyons)

Urban canyon effect



Positioning accuracy (typical values)

1 Stand-alone GPS

- No additional corrections.
- Typical (SPS) accuracy: **3 – 10 m.**



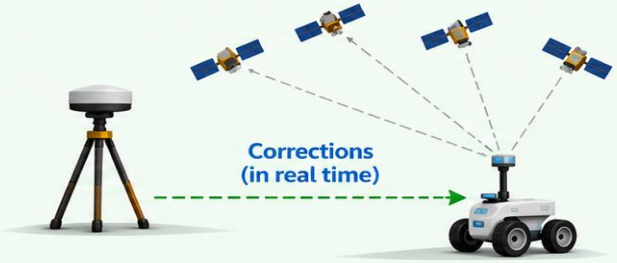
2 Differential GPS (DGPS)

- Use a nearby reference station for corrections.
- Typical accuracy: **0.5 – 3 m.**



3 RTK GPS

- Carrier-phase measurements and real-time corrections (reference station).
- Typical accuracy: **1 – 5 cm.**



Range sensors: Sonar



Early ultrasonic ranging systems were used by **Polaroid** for camera autofocus (in the 1970s, e.g., the SX-70).

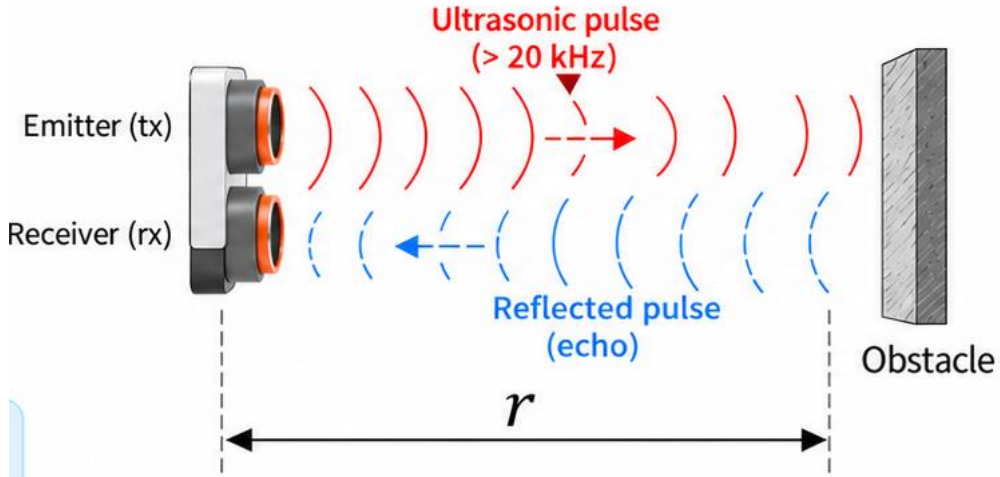
Working principle:

1. The emitter sends a short **ultrasonic pulse** (> 20 kHz).
2. The pulse is **reflected** by an obstacle and detected by the receiver.
3. The **round-trip time Δt** is measured to estimate the distance (Time of Flight, ToF).

$$r = \frac{c_s \, \Delta t}{2}$$

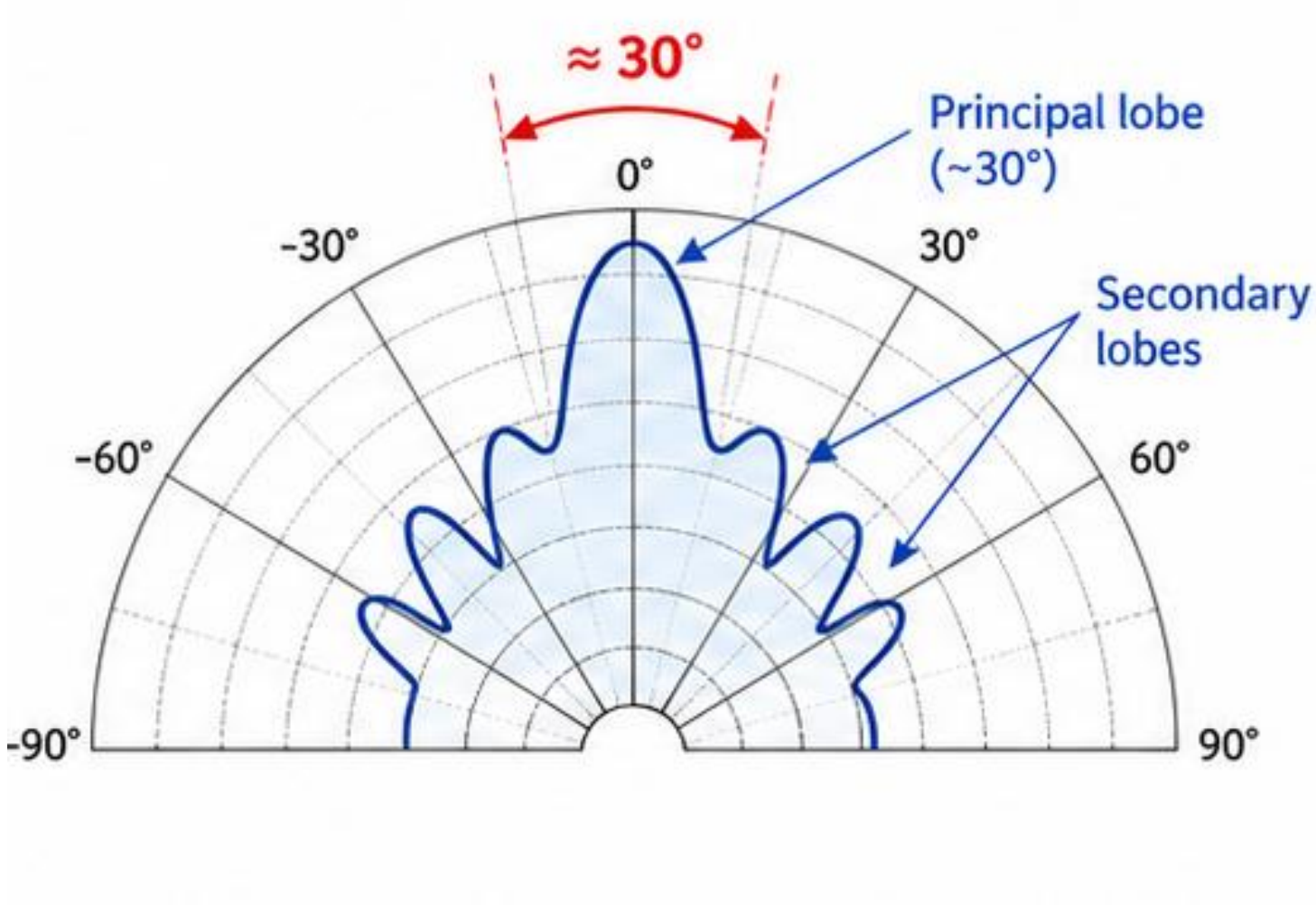
c_s : speed of sound in air (≈ 343 m/s at 20 °C).

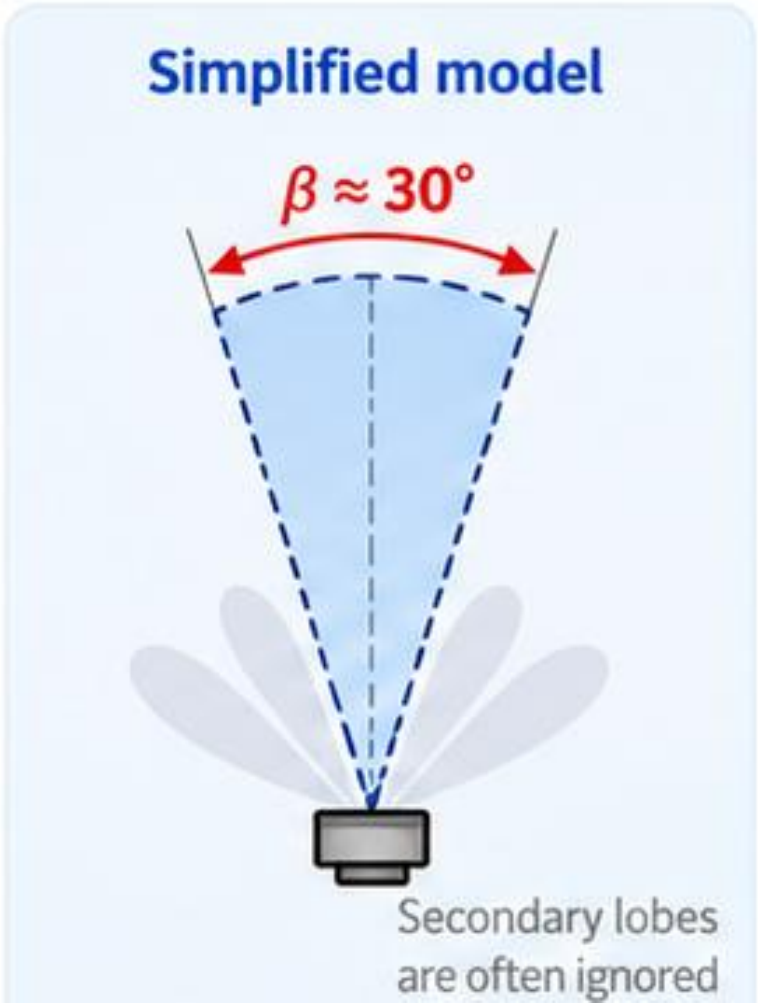




Range sensors: Sonar

- The effective beam width is about 30°, but there are secondary lobes.

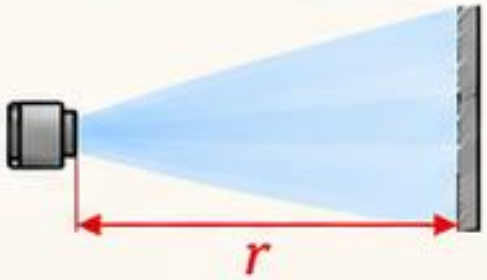




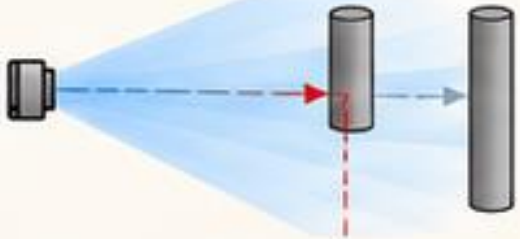
Range sensors: Sonar

Typical measurement situations:

Accurate distance

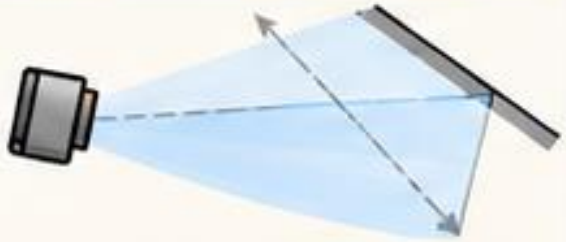


Low lateral resolution



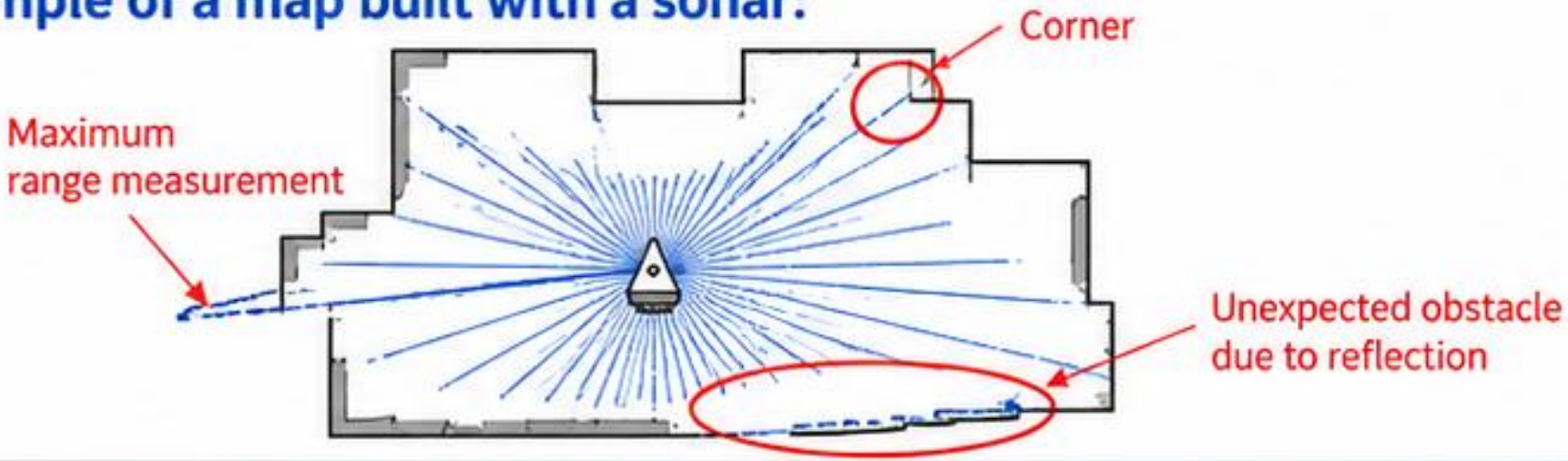
Closest object in the beam is detected.

Specular reflection / missed echo



Smooth surfaces may reflect the sound away from the sensor (no echo).

Example of a map built with a sonar:



Range sensors: Sonar

Key features

Low cost
Inexpensive and widely available.

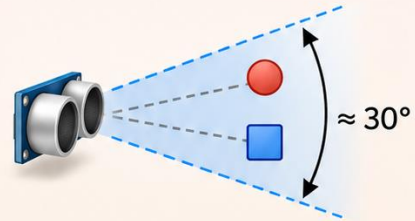
Compact
Small size and easy to integrate.

Low power
Low power consumption.

Short/medium range
Typically up to 2–4 m (depending on the sensor).

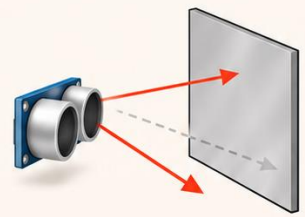
Useful for obstacle detection
Well suited for simple navigation and collision avoidance.

Limitations



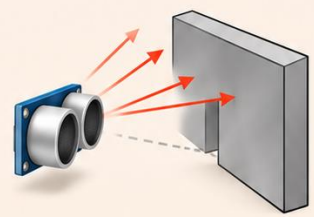
Poor angular resolution

- Wide beam (≈ 30°).
- Objects separated laterally can be at the same distance and cannot be distinguished.



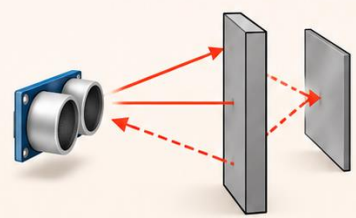
Sensitive to surface type

- Specular (smooth) surfaces reflect the sound away from the sensor.
- This can result in weak or no echo (missed detection).



Corners can be missed

- Open corners do not reflect enough sound back to the sensor.
- This can result in weak or missing measurements.

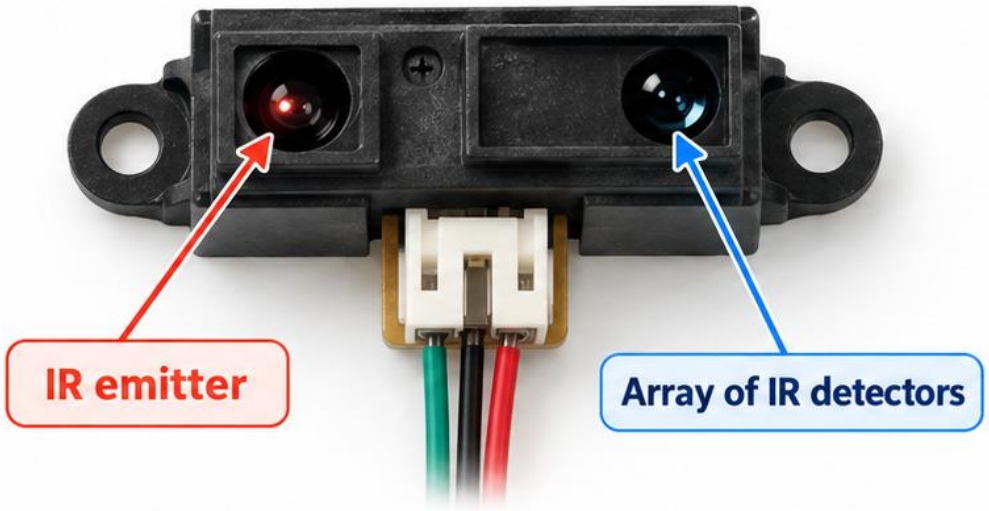


Multiple reflections

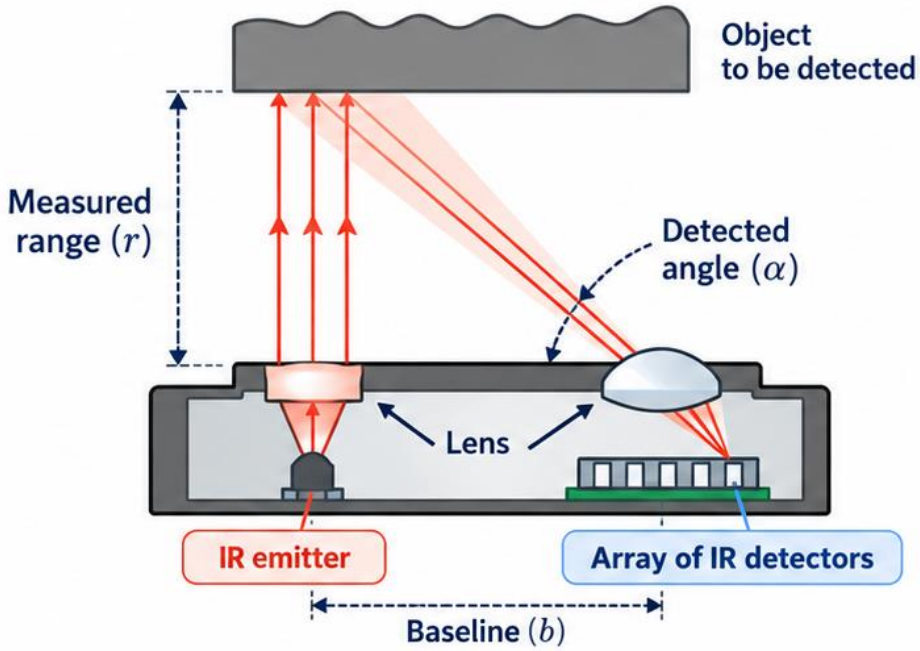
- Sound can bounce off several surfaces before returning to the sensor.
- This can produce spurious echoes and erroneous distance measurements.

Range sensors: Infrared

Based on triangulation of light



Operating principle



Distance is computed from the detected angle

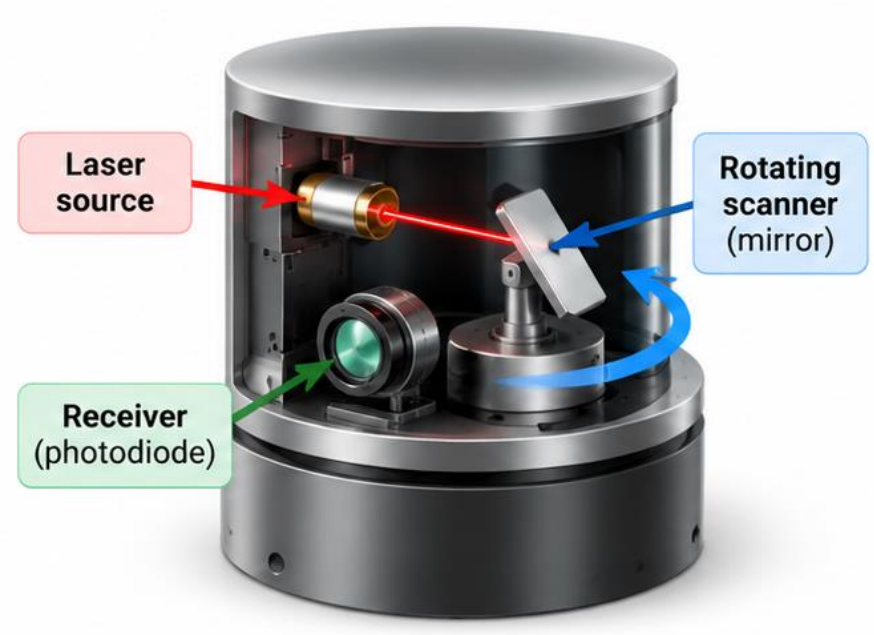
Features

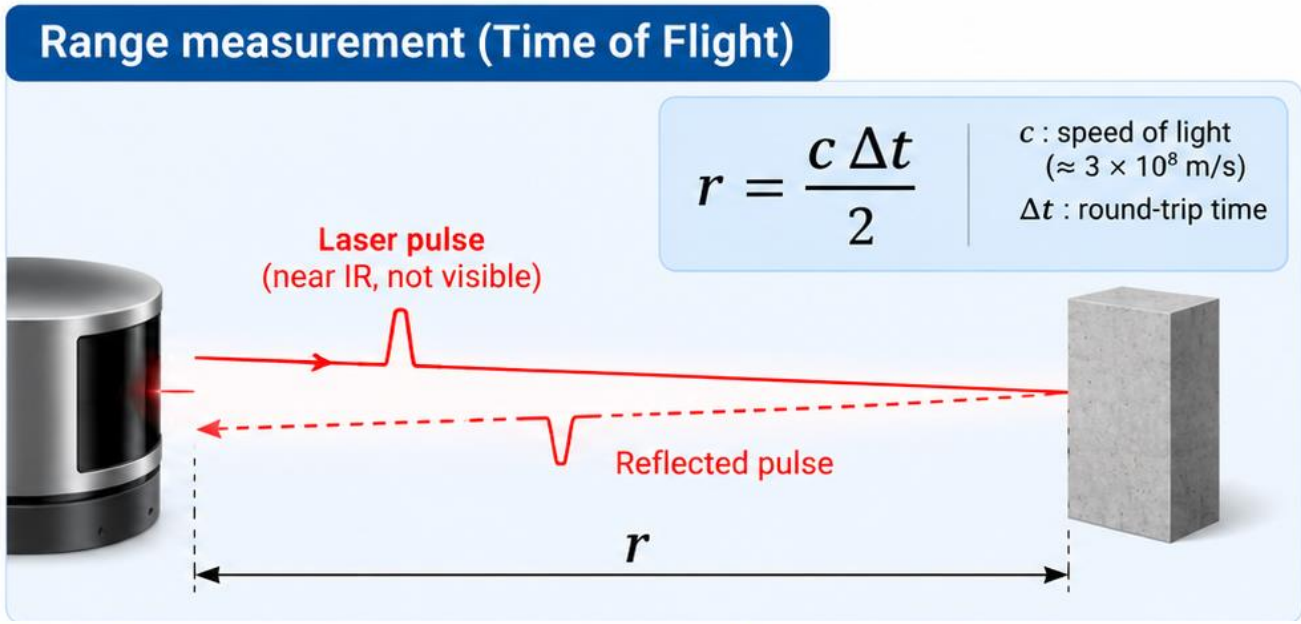
Small, cheap, low power consumption

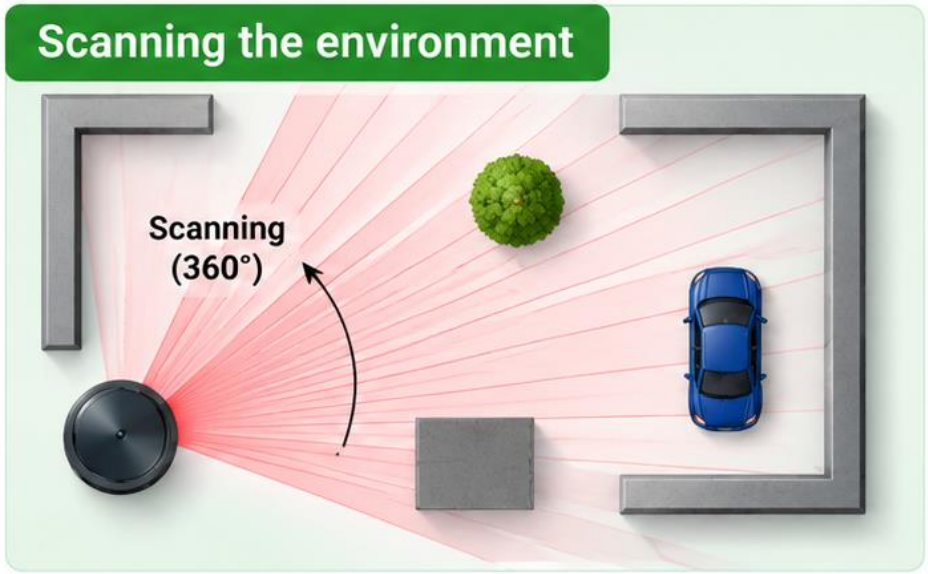
Range up to about **2 m**; **good angular resolution** (narrower beam than sonar)

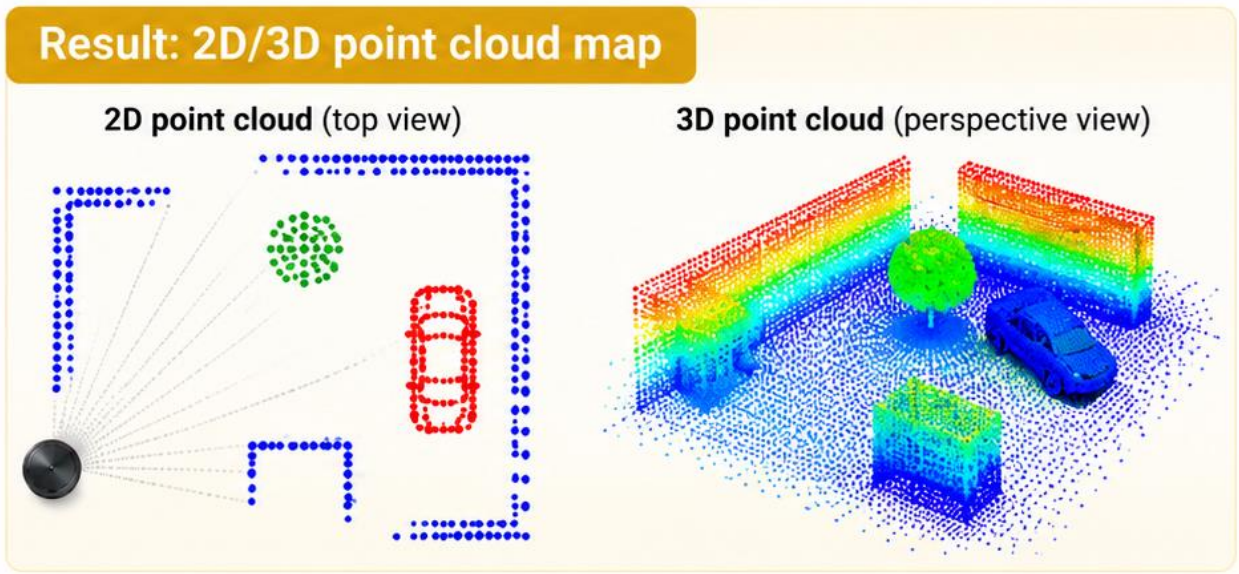
Sensitive to **transparent, glossy, and black objects**, and to **sunlight**

Range sensors: Laser scanner (LIDARS)









Range sensors: Laser scanner (LIDARS)

Performance characteristics

Range
From a few metres to 200 m or more, depending on the sensor and target reflectivity.

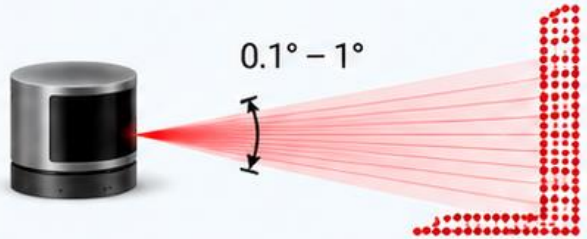
Ranging accuracy
Typically centimetre-level (~ 1–5 cm) for many sensors.

Angular resolution
Narrow beam (typically 0.1°–1°) allows detailed spatial measurement.

Cost
Generally more expensive than ultrasonic sensors, but rapidly decreasing due to the automotive industry and mass production.

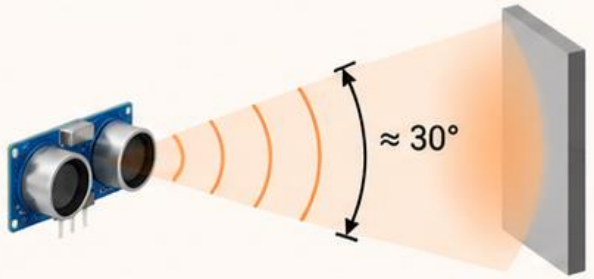
Comparison with ultrasonic sensors

LiDAR (typical)



- Narrow beam (0.1° – 1°).
- High angular resolution.
- Detailed 2D/3D point clouds.
- Long range (up to 200 m or more).
- Centimetre-level accuracy.

Ultrasonic (typical)



- Wide beam (≈ 30°).
- Poor angular resolution.
- Provides only distance information within the beam.
- Short/medium range (typically ≤ 4 m).
- Lower cost.

Limitations

Environmental conditions
Performance can degrade in rain, fog, dust or smoke due to scattering and attenuation of the laser beam.



Low reflectivity surfaces
Dark objects or absorbing materials (e.g., black carpet, glass) may return weak or no signal.



Specular reflections
Shiny or reflective surfaces (e.g., mirrors, polished metal) can deflect the beam away from the sensor.



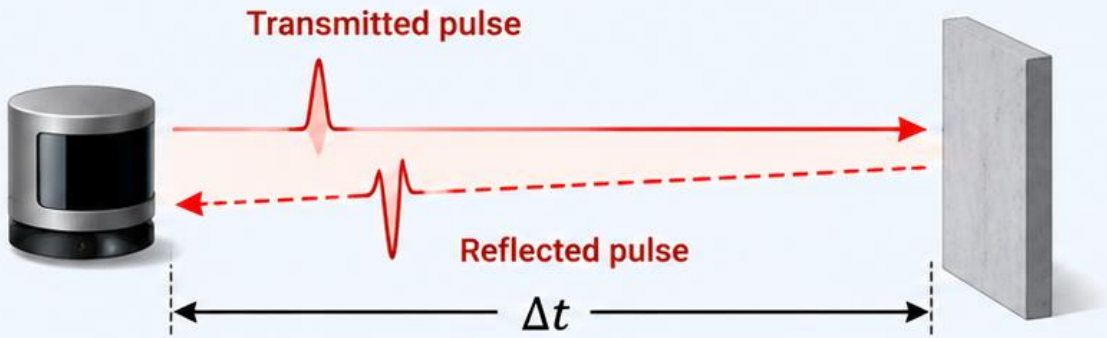
Moving objects
Motion during scanning can cause distortions in the point cloud.



LiDAR ranging techniques

Pulsed time-of-flight (ToF)

Transmit short laser pulses and measure the round-trip time Δt.



$$r = \frac{c\,\Delta t}{2}$$

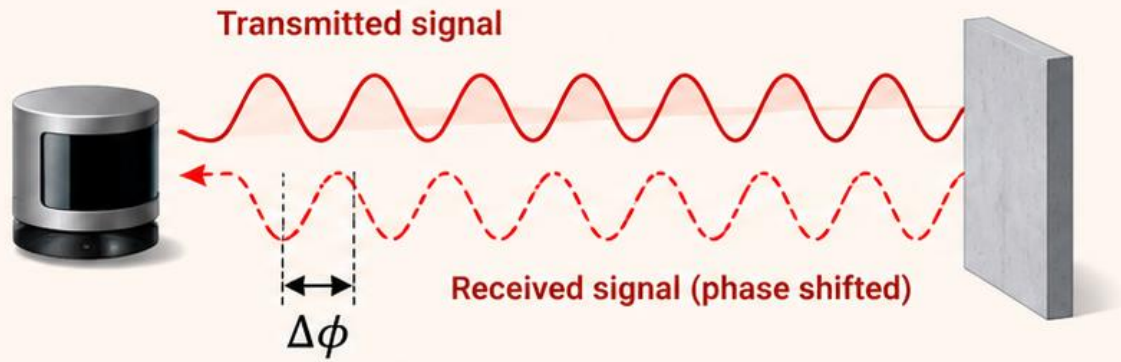
c : speed of light ($\approx 3 \times 10^8$ m/s)

Δt : round-trip time

- Direct measurement of the propagation time.
- Well suited for long-range measurements (up to hundreds of metres).
- Used in many autonomous driving LIDAR sensors.

Phase shift (amplitude-modulated)

Transmit continuously modulated laser light and measure the phase shift $\Delta\phi$ of the received signal.



$$r = \frac{c\,\Delta\phi}{4\pi f_m}$$

c : speed of light ($\approx 3 \times 10^8$ m/s)

$\Delta\phi$: phase shift (radians)

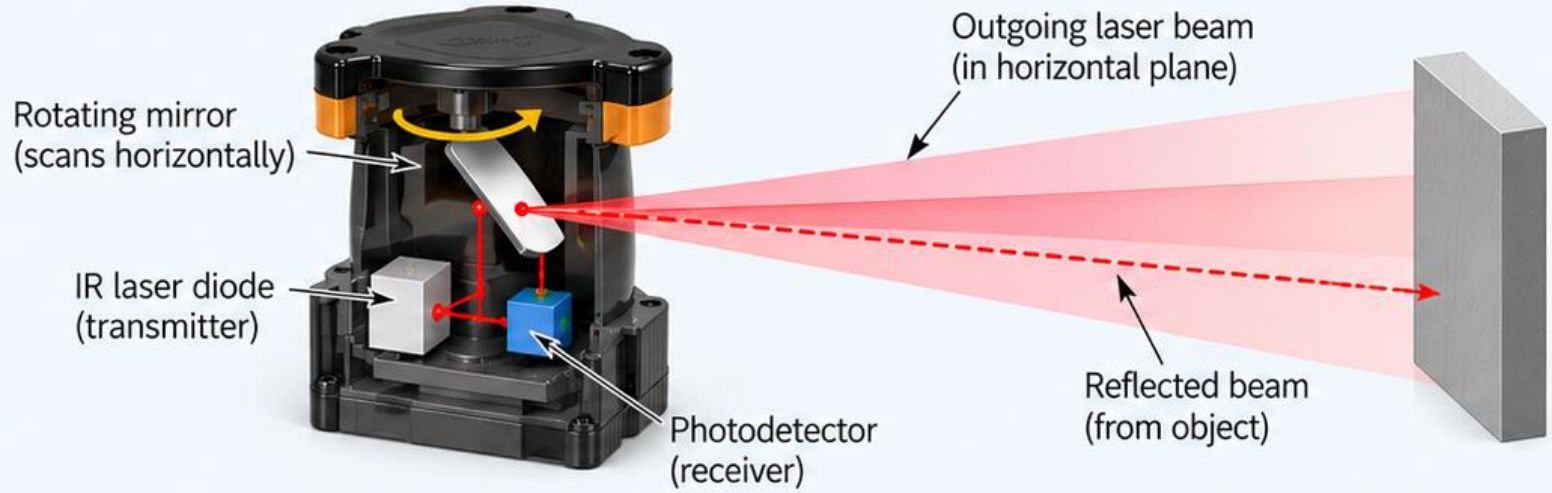
f_m : modulation frequency

- Indirect measurement of the propagation time.
- High precision at short/medium range.
- Range is ambiguous beyond $\frac{c}{2\,f_m}$ (requires additional techniques to extend the unambiguous range).
- Used in many short/medium-range LiDAR sensors (e.g., indoor, robotics).

Example: 2D LiDAR for mobile robots (Hokuyo UTM-30LX)



Internal structure and scanning principle



Main specifications (Hokuyo UTM-30LX)

Detection range
30 m (typical, max.)

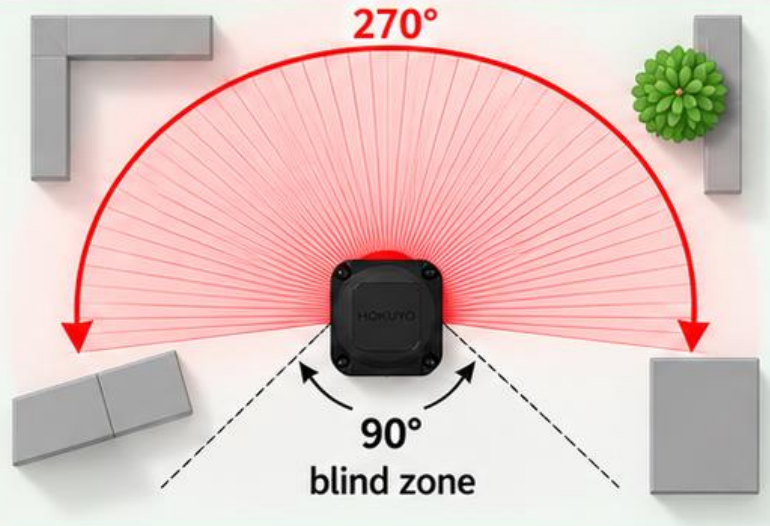
Field of view (FOV)
270°

Suitable environment
Outdoor use

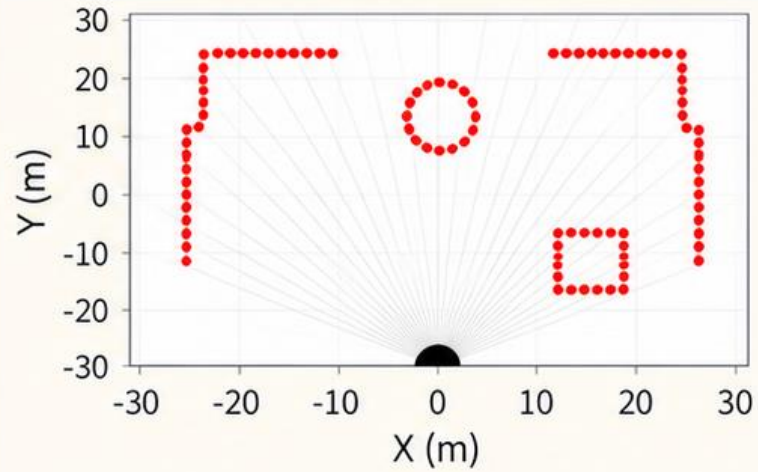
Size
W 60 × D 60 × H 87 mm

Weight
370 g

Horizontal scanning (top view)



Example scan result

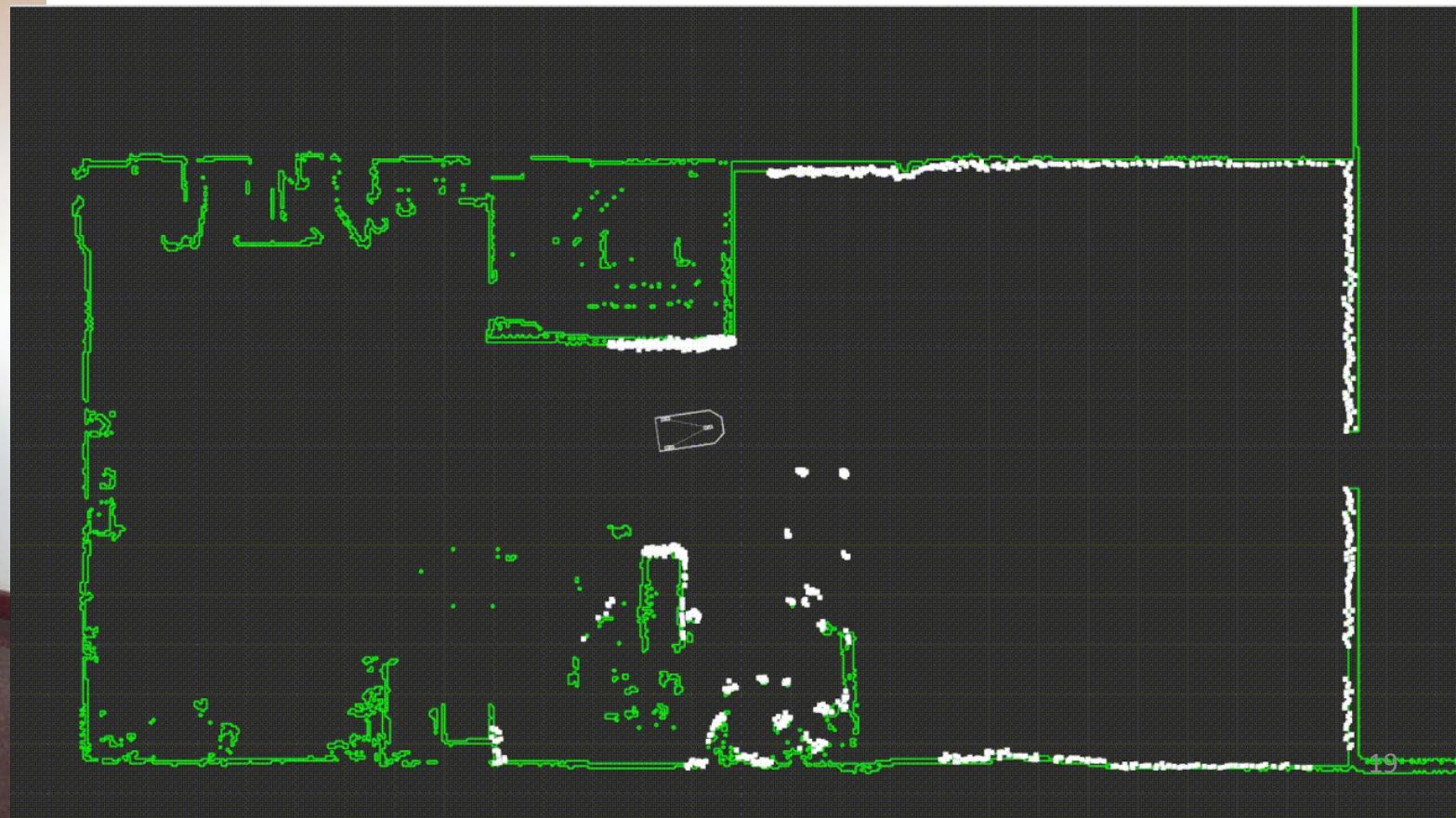






Hokuyo Radial
Laser Scanner

Example of a scan(270º FoV)



4D LiDAR in modern robots (e.g., Unitree)



Unitree 4D LiDAR
(e.g., L1 / L2)

What does 4D mean?

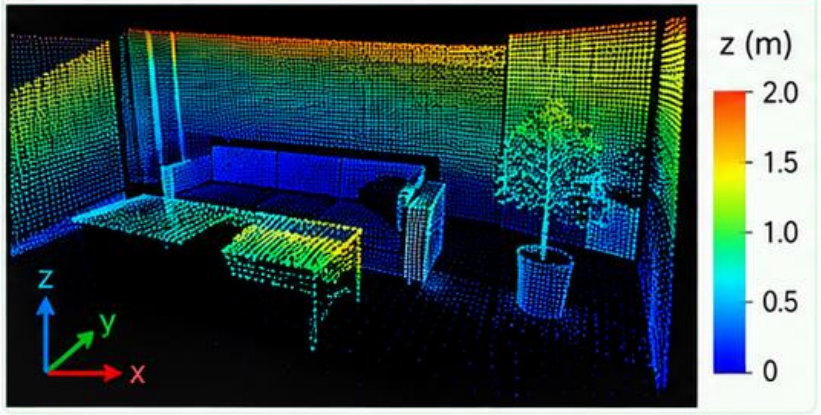
Each measurement provides 3D position and an additional **attribute:** the reflectivity (intensity) of the returned signal.

4D point:

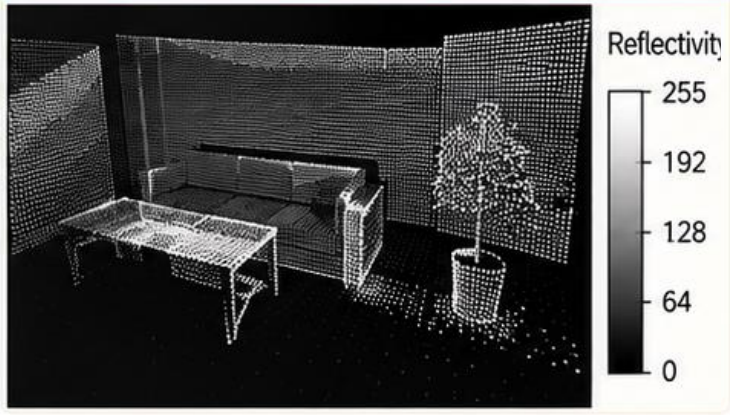
$$p_i = (\underbrace{x_i, y_i, z_i}_{\text{3D position (geometry)}}, \underbrace{g_i}_{\text{Reflectivity (grayscale)}})$$

- The fourth dimension is not an additional spatial coordinate.
- It is the intensity or reflectivity of the returned laser signal.

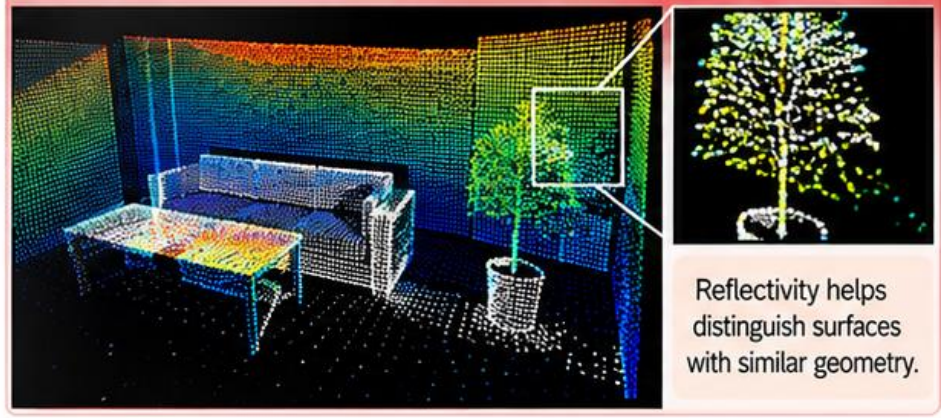
3D point cloud (geometry)



Reflectivity (grayscale)



4D point cloud (geometry + reflectivity)



Reflectivity helps distinguish surfaces with similar geometry.

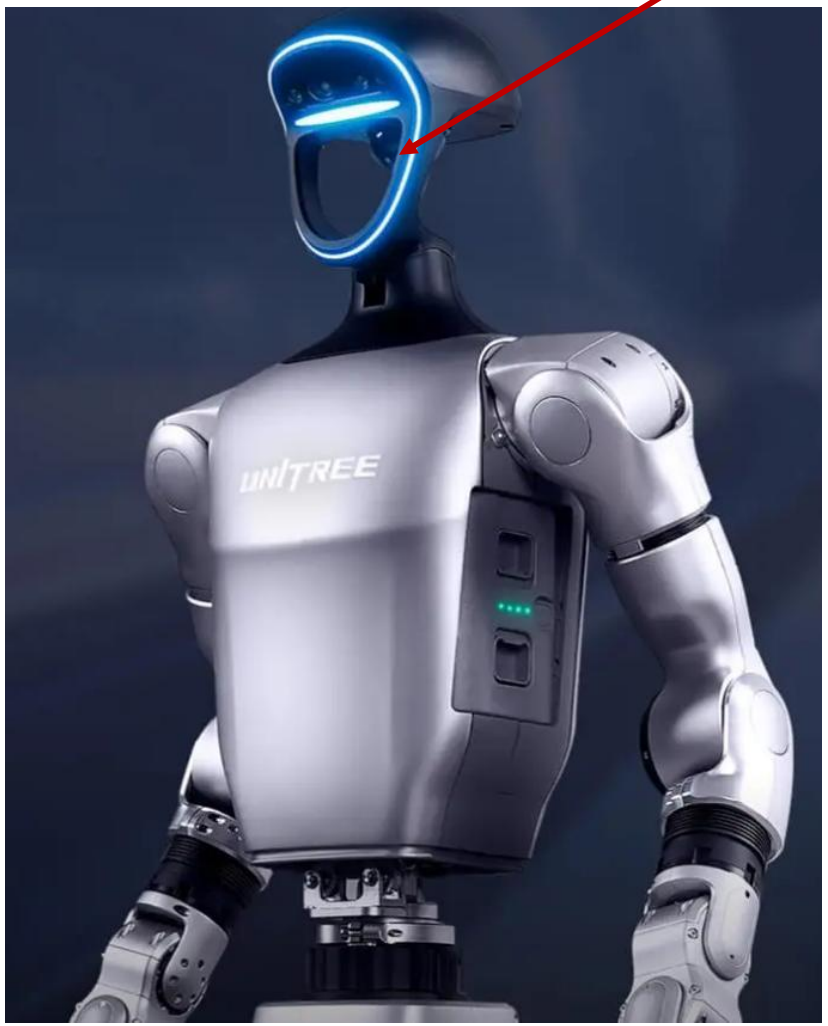
Example (Unitree L2)

Horizontal FOV	Vertical FOV	Range	Points per second	4D point: (x, y, z, reflectivity)
360°	96°	~ 120 m (max)	~ 1.2 million	Reflectivity: 0 – 255 (grayscale)

Range sensors: 3D Lidar

… and for robots

Lidar 3D



Unitree android



Unitree - GO 2 AIR DOG Robot

Range sensors: 3D Lidar

Getting very popular in autonomous cars (for example **Waymo Car**)

Waymo robotaxi
by Google



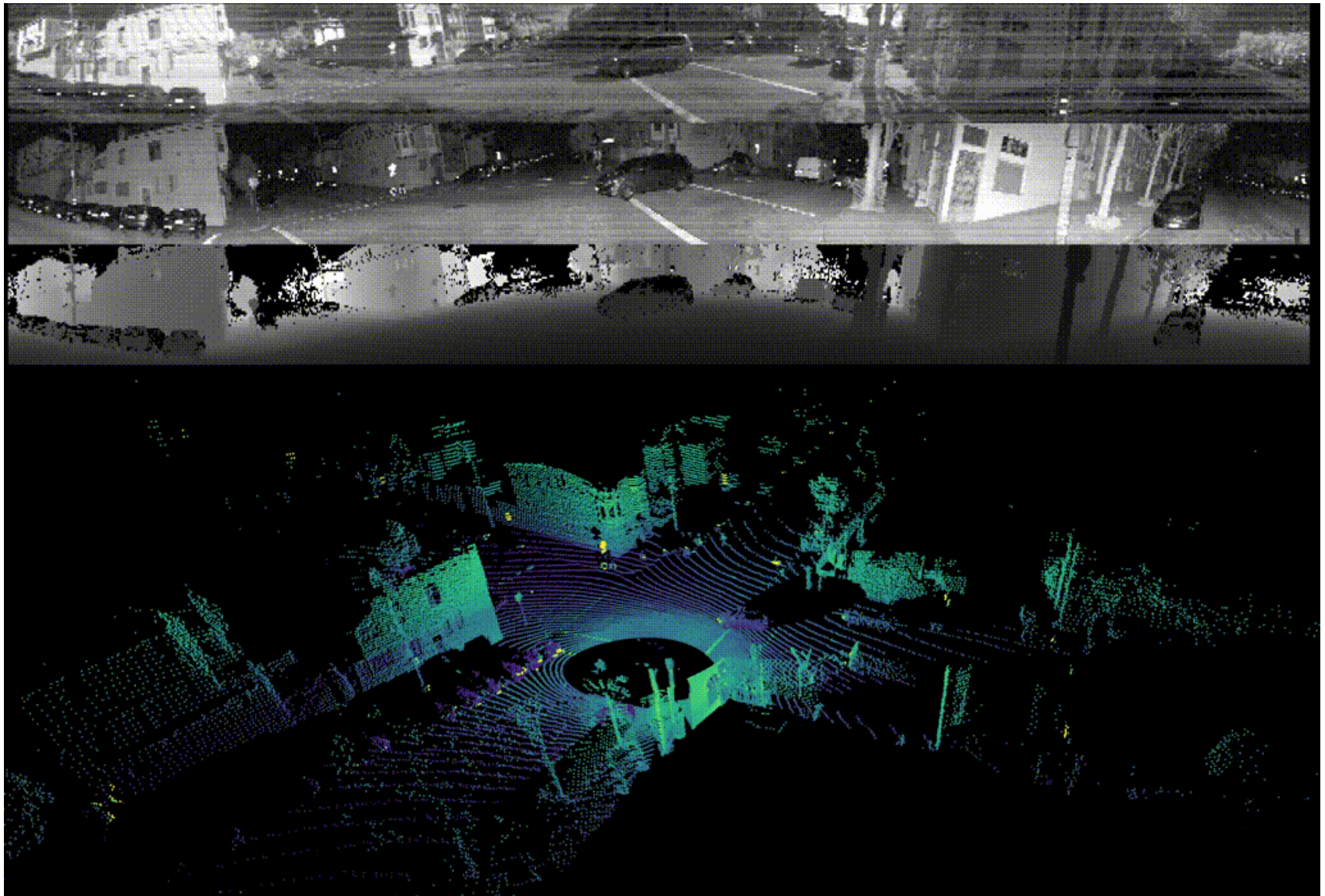
- 360° LiDAR on the roof with a range of +300m
- four short-range perimeter LiDARs around the vehicle

Range sensors: 3D Lidar

Getting cheaper and more used everyday



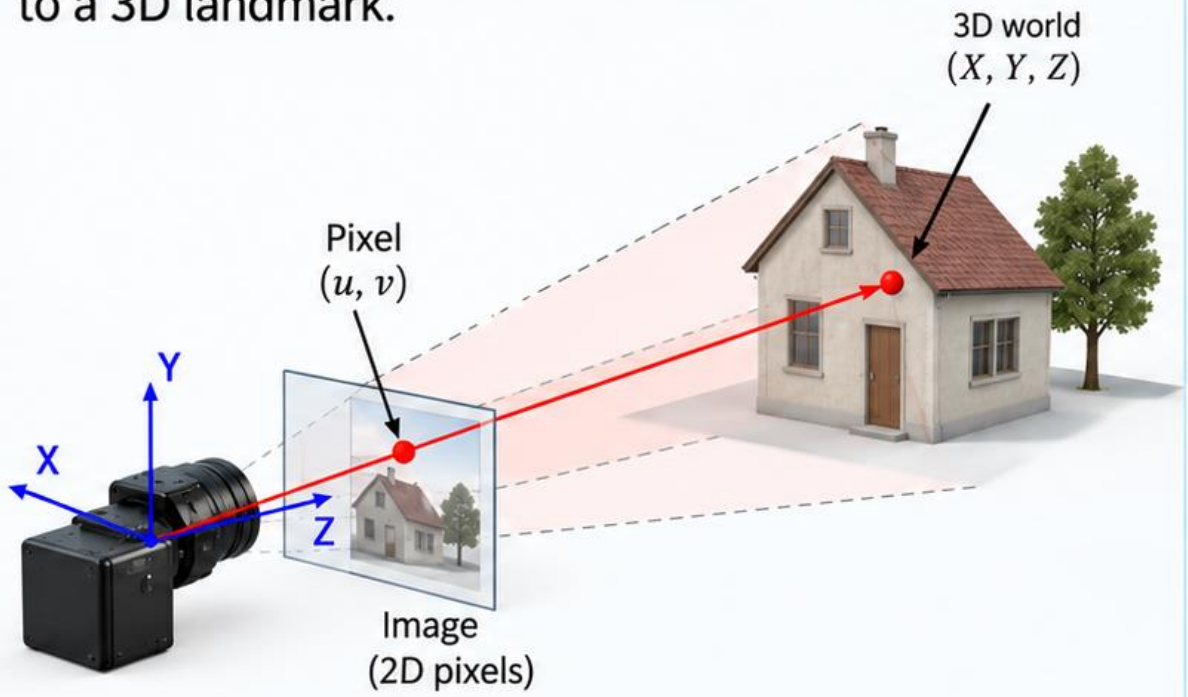
Ouster OS1 3D Lidar



Cameras

Single camera

A **single camera** provides **only bearing information** to a 3D landmark.

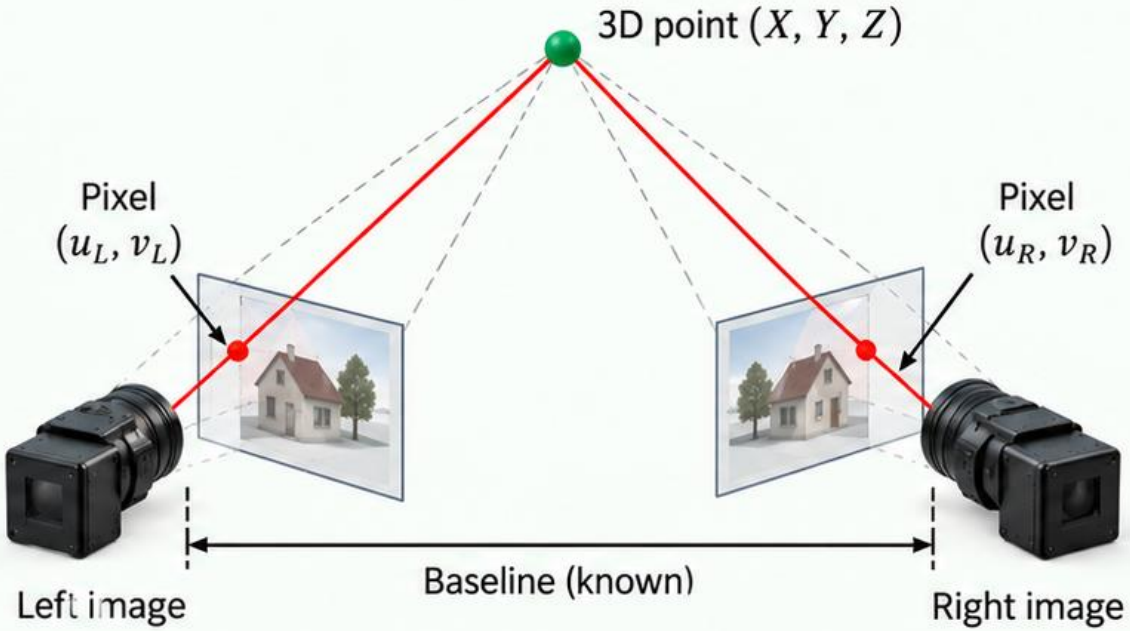


Result:

A point (pixel) in the image gives a **line in space** (line-of-sight of the pixel).

Stereo cameras

Stereo cameras use two views to provide **range-and-bearing** measurements.



Result:

A 3D point is obtained through **triangulation** of the line-of-sight of two pixels.

Cameras

RGB Cameras in mobile robotics: MAPIR research



Human-robot interaction

RGB-D Cameras

RGB-D camera

- Provides a **color RGB image** and a **depth image** (per pixel).
- In contrast to 3D laser scanners:
  - No moving parts (no rotating mirrors).
  - Dense depth map (full image).
  - Typically shorter range (< 4 m).
- Each pixel provides both appearance (color) and distance (range).



Intel RealSense D455
(example RGB-D camera)







Color image
Standard RGB image
(e.g., 640×480 or higher)

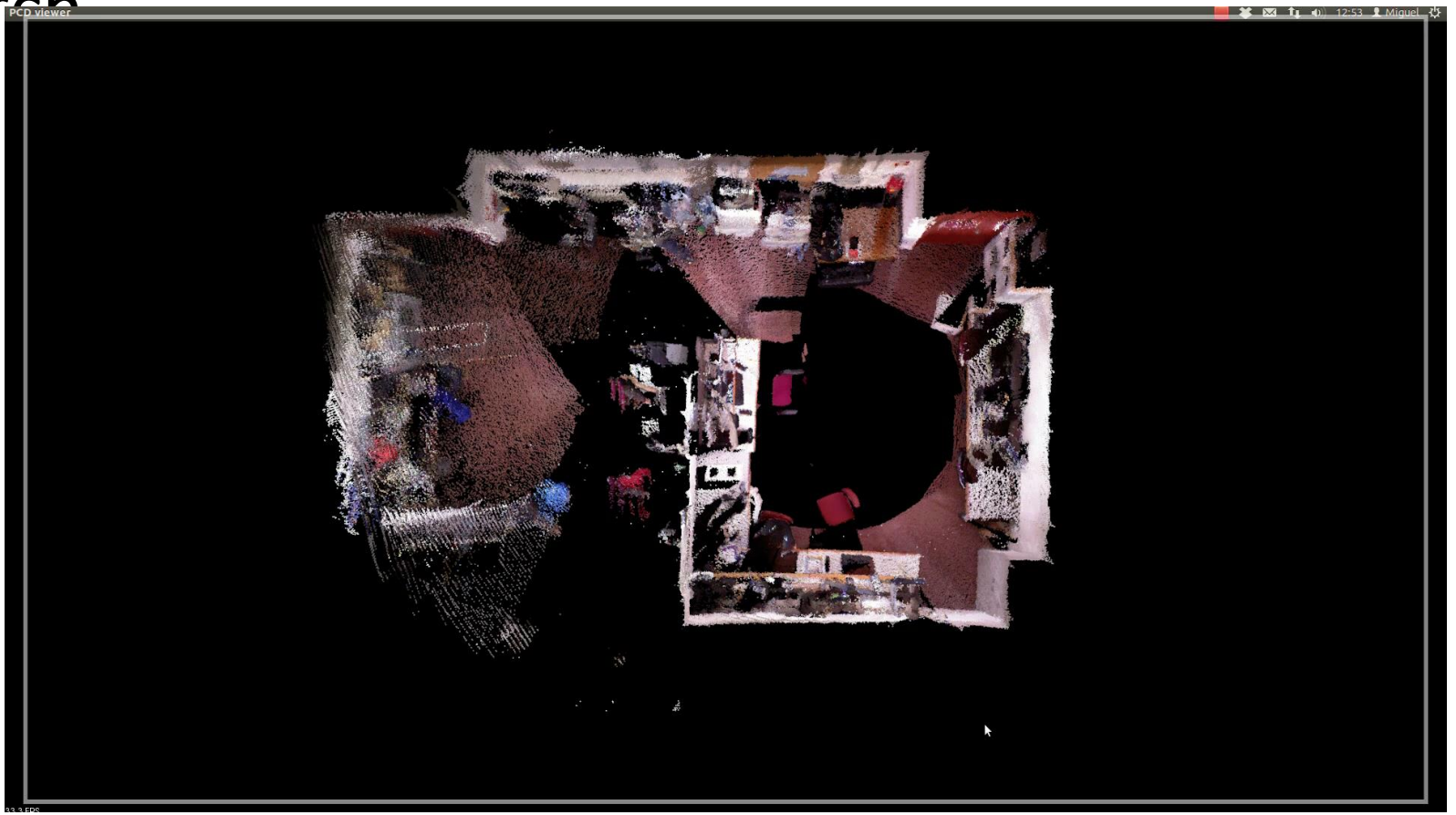
Depth image
Depth value for each pixel
(dense per-pixel map)

No moving parts
Solid-state sensor
(no rotating mirrors)

Typical range
~ 0.2 – 4 m
(indoor use)

RGB-D Cameras

RGB-D Cameras in mobile robotics: MAPIR research



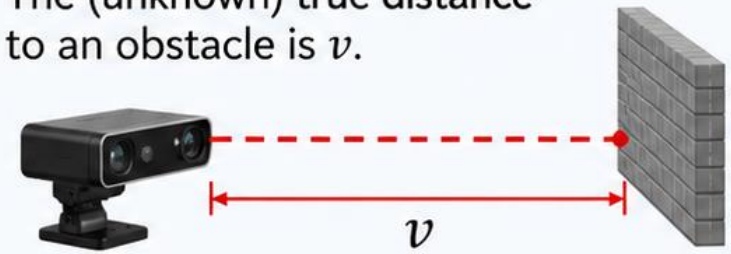
Localization&mapping (SLAM)

Probabilistic Sensor Models

Real sensors do not deliver the exact truth of the quantities they are measuring, but a perturbed version: truth + error.

1 True distance

The (unknown) true distance to an obstacle is v.



2 Repeated measurements

If we repeat the **measurement**, we obtain different values:

$$z_1, \quad z_2, \quad z_3, \quad \dots$$

3 Probabilistic model

These measurements are **realizations** of a random variable

$$Z \sim p(z \mid v)$$

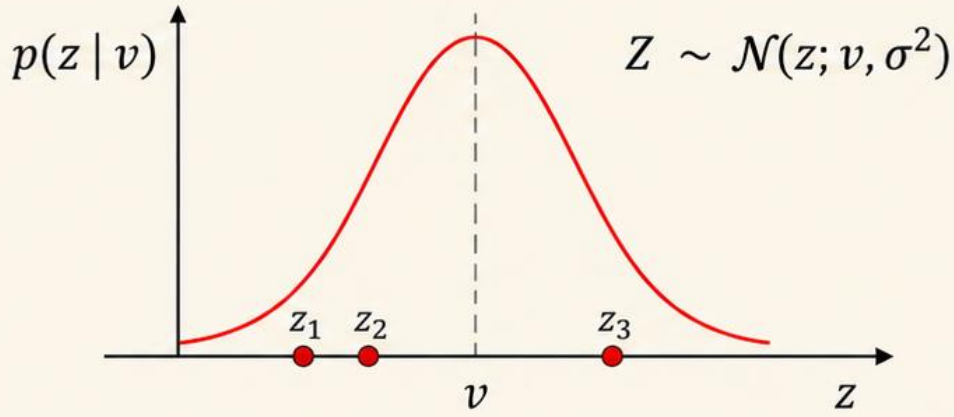
Example: Gaussian measurement model

The sensor error is often modeled as Gaussian:

$$p(z \mid v) = \mathcal{N}(z; v, \sigma^2)$$

- Z : random variable (measurement)
- z : one particular measurement (realization)
- v : true value (unknown)
- σ^2 : measurement variance

Distribution of measurements



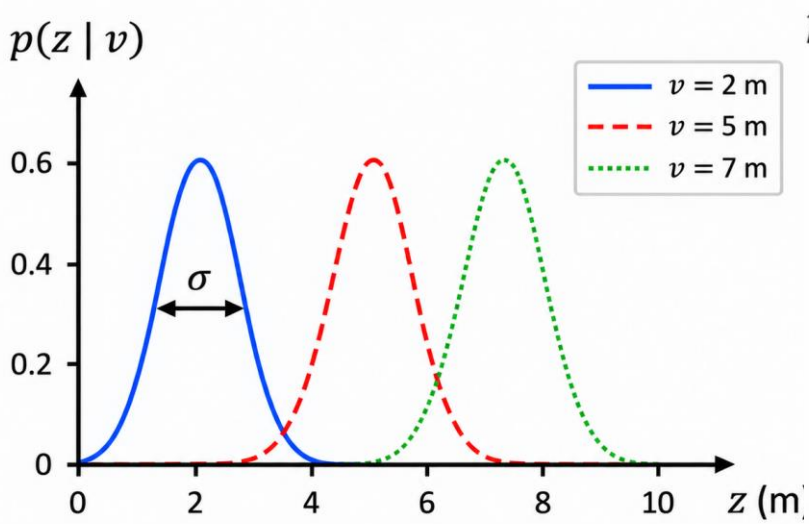
Probabilistic Sensor Models

Examples of Gaussian error sensors in 1D

1. Constant uncertainty

$$Z \mid v \sim \mathcal{N}(v, \sigma^2)$$

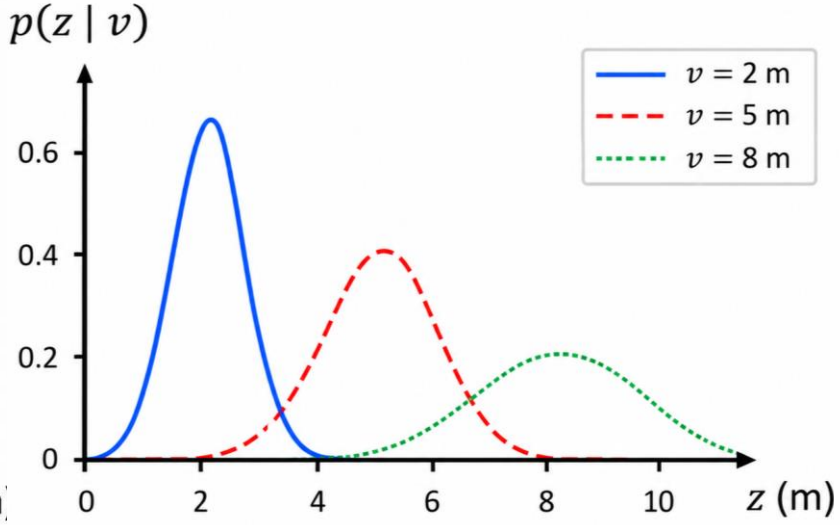
σ constant



2. Uncertainty increases with distance

$$Z \mid v \sim \mathcal{N}(v, \sigma^2(v))$$

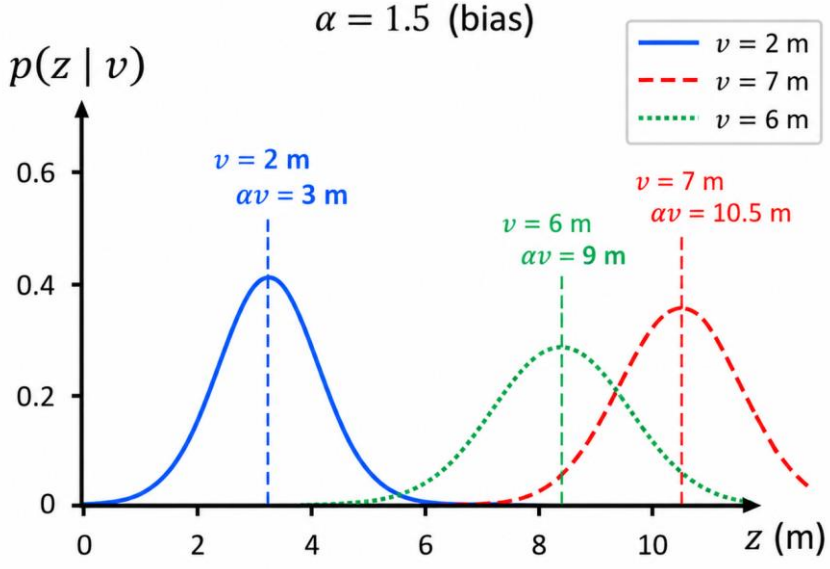
$\sigma(v)$ increasing with v



3. Systematic bias (scale error)

$$Z \mid v \sim \mathcal{N}(\alpha v, \sigma^2(v))$$

$\alpha = 1.5$ (bias)



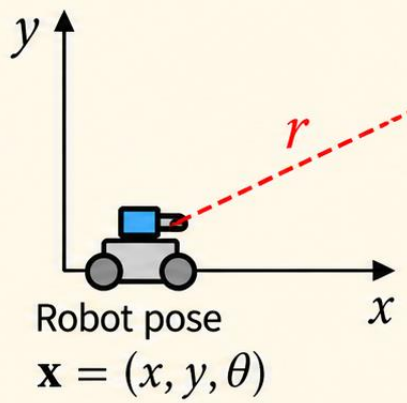
Probabilistic Sensor Models

1 Ideal measurement model

The ideal (noise–free) measurement vector $\mathbf{v}$ depends on the sensor pose $\mathbf{x}$ and the environment (map $\mathbf{m}$)):

$$\mathbf{v} = \mathbf{h}(\mathbf{x}, \mathbf{m})$$

Example: range measurement to a landmark



Range function:

$$v = h(\mathbf{x}, \mathbf{m}) = r = \sqrt{(m_x - x)^2 + (m_y - y)^2}$$

2 Probabilistic sensor model

The sensor does not give the ideal value v, but a random measurement $\mathbf{Z}$ due to noise:

$$p(\mathbf{z} \mid \mathbf{x}, \mathbf{m}) = p(\mathbf{z} \mid v = \mathbf{h}(\mathbf{x}, \mathbf{m}))$$

Actual observation/ realization of Z

map of the environment

sensor pose

3 Gaussian measurement model

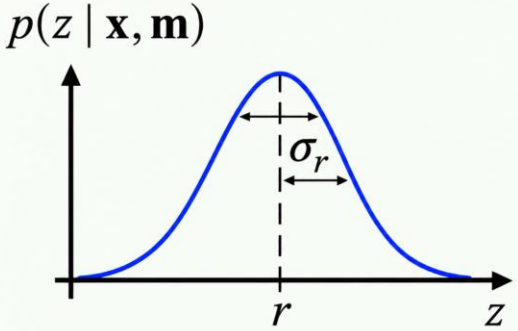
We assume that the measurement is the ideal value plus Gaussian noise:

$$\mathbf{Z} = h(\mathbf{x}, \mathbf{m}) + \mathbf{W}, \quad \mathbf{W} \sim \mathcal{N}(0, Q) \quad \Rightarrow \quad p(z \mid \mathbf{x}, \mathbf{m}) = \mathcal{N}(\mathbf{z}; \mathbf{h}(\mathbf{x}, \mathbf{m}), Q)$$

Example (range)

Measured range to a landmark:

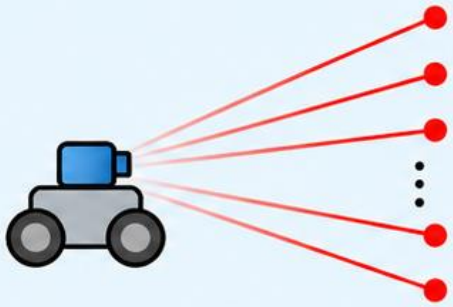
$$Z = r + W, \quad W \sim \mathcal{N}(0, \sigma_r^2)$$



Probabilistic Sensor Models

1 One observation = many measurements

- An observation $\mathbf{z}$ usually contains several measurements.
- They may come from one device or from multiple sensors.

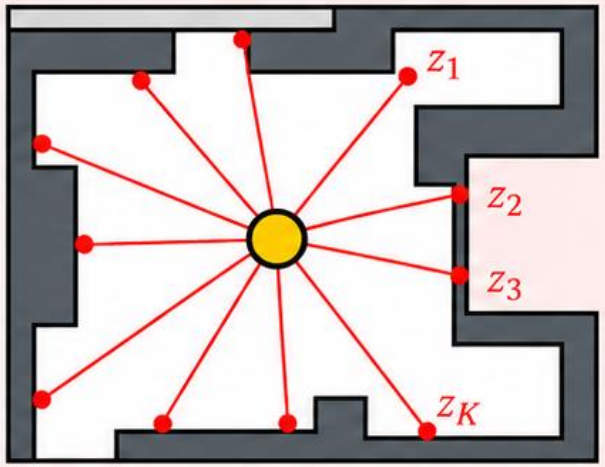


$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ z_3 \\ \vdots \\ z_K \end{bmatrix}$$

K measurements

2 Example: a laser scan

A single scan provides many range measurements z_k.



Top view

3 Conditional independence assumption

Given the robot pose $\mathbf{x}$ and map $\boldsymbol{m}$, the individual measurements are assumed conditionally independent.

$$p(\mathbf{z} \mid \mathbf{x}, \mathbf{m}) = \prod_{k=1}^{K} p(z_k \mid \mathbf{x}, \mathbf{m})$$

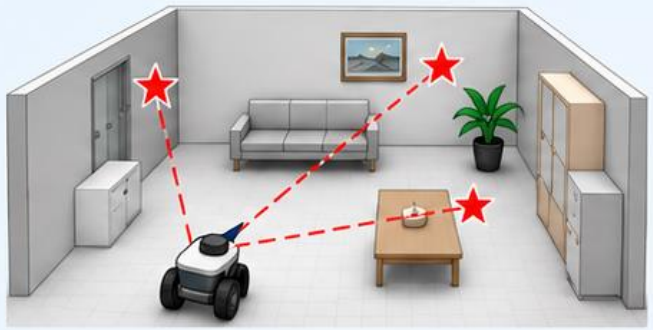
Recall: z_1 and z_2 are conditionally independent given $\mathbf{x}, \boldsymbol{m}$ if $p(z_1, z_2 \mid \mathbf{x}, \mathbf{m}) = p(z_1 \mid \mathbf{x}, \mathbf{m}) p(z_2 \mid \mathbf{x}, \mathbf{m})$.

The joint likelihood is approximated as the product of the individual likelihoods.

For a sensor model $p(\mathbf{z}|\mathbf{x}, \mathbf{m})$ we need a map $(\mathbf{m} = \mathcal{M})$

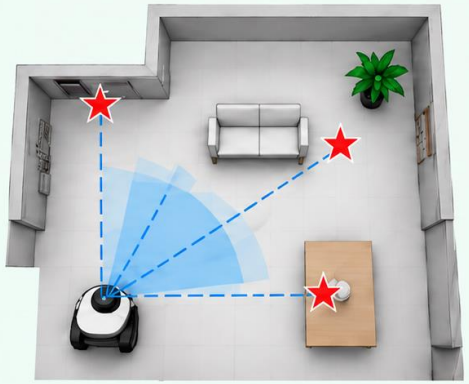
What is a map?

A **geometric map** $\mathcal{M}$ is a representation of the spatial structure of the environment.



Localization

Estimate the robot pose using sensor observations and the map.



Role of the map

Predict expected observations at a candidate pose and evaluate their likelihood.

Predict measurement

$$\hat{z} = h(\mathbf{x}, \mathcal{M})$$

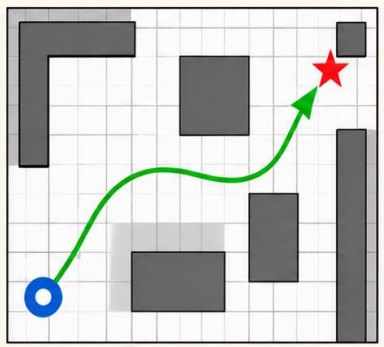
Measurement likelihood

$$p(z \mid \mathbf{x}, \mathcal{M}) = p(z \mid h(\mathbf{x}, \mathcal{M}))$$

Probability of obtaining measurement z given pose $\mathbf{x}$ and map $\mathcal{M}$.

Navigation and planning

Use the map to plan collision-free paths to a goal and to avoid obstacles.



Role of the map

Provides the environment geometry (obstacles, free space and goal) to plan feasible and safe paths.

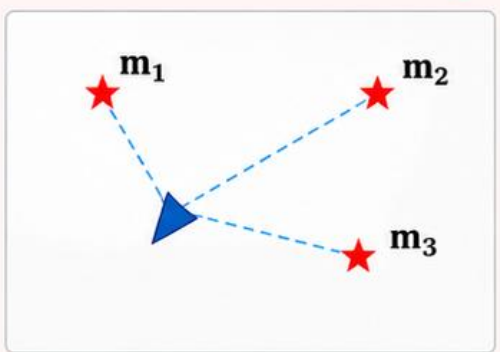
Plan a path

$$\pi = \text{plan}(\mathbf{x}_{\text{start}}, \mathcal{M})$$

- Avoid collisions (obstacles).
- Reach the goal.
- Optimize a criterion (e.g. path length, safety, …).

Two common geometric map representations

Landmark-based map

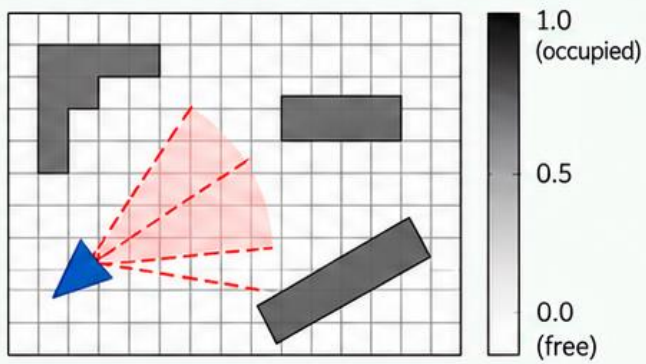


Representation:

$$\mathcal{M} = \{\mathbf{m}_1, \mathbf{m}_2, \ldots, \mathbf{m}_N\}$$

- Each landmark $\mathbf{m}_i$ includes the position $(\mathbf{x}, \mathbf{y})$ and a descriptor (e.g., visual features).
- Typically used with landmark detections from cameras, beacons, etc.
- **Sensor models:**
  - Bearing (angle)
  - Range (distance)
  - Range and bearing

Occupancy grid map



Representation:

$$\mathcal{M} = \{m_{ij}\}$$

- Each cell m_{ij} stores an occupancy probability $p(m_{ij}) = P(\text{occupied})$.
- Commonly used with LiDAR scanners and depth sensors.
- **Sensor models:**
  - Beam model
  - Likelihood-field model

Likelihood vs. Conditional Probability

Bayes' theorem (localization)

$$\underbrace{p(x \mid z, m)}_{\text{posterior}} \propto \underbrace{p(z \mid x, m)}_{\text{likelihood}} \underbrace{p(x \mid m)}_{\text{prior}}$$

posterior (distribution over x) — likelihood (measurement model) — prior (previous belief)

Conditional probability (PDF) $p(z \mid x, m)$

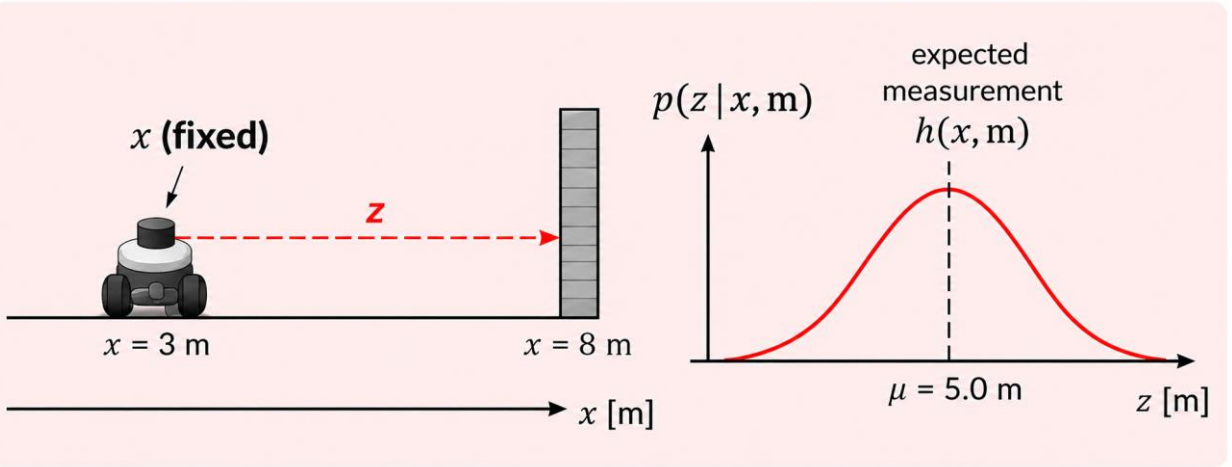
- For a fixed robot pose x and map m, **z is a random variable.**
- Describes the distribution of possible measurements the sensor can produce.
- It is a proper probability distribution:

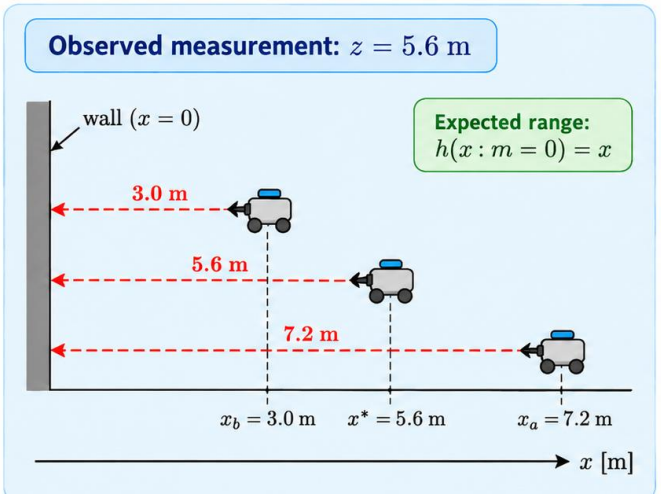
$$\int_{-\infty}^{\infty} p(z \mid x, m) dz = 1$$

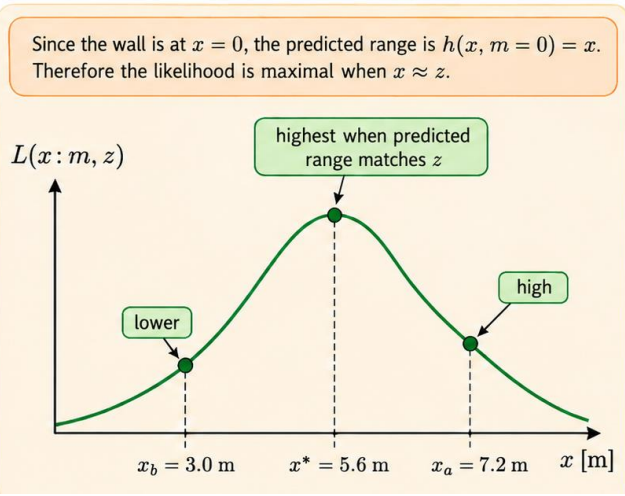
Likelihood (as a function of x) $p(z \mid x, m) = \mathcal{L}(x; m, z)$

- For fixed measurement z and map m, **x is the variable.**
- Measures how **compatible** each candidate robot pose x is with the observed measurement z.
- It is **not** a probability distribution over x:

$$\int_{-\infty}^{\infty} \mathcal{L}(x; m, z) dx \neq 1$$





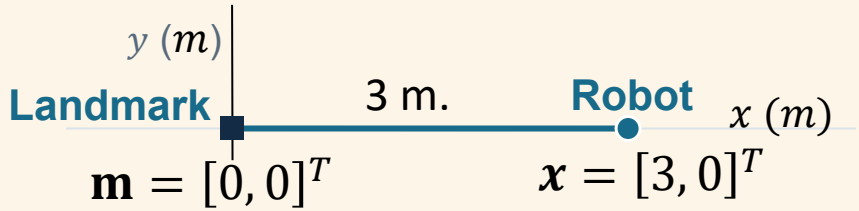


Example: A range sensor measuring **distance r to a Landmark in 2D**

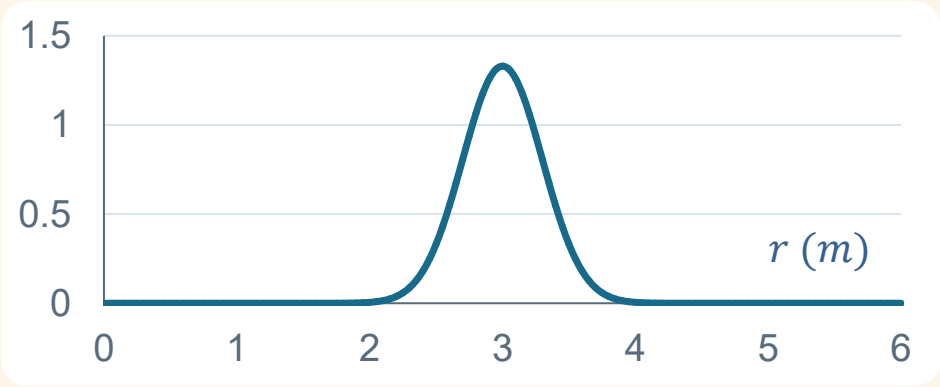
$$r = \|\boldsymbol{x} - \boldsymbol{m}\| + w \quad w \sim \mathcal{N}(0, \sigma^2) \rightarrow p(r|\mathbf{x}, \mathbf{m}) = K \cdot \exp\{-\frac{1}{2}\frac{(\|\mathbf{x}-\mathbf{m}\|-r)^2}{\sigma^2}\}$$

Conditional probability

Possible r for a given $\mathbf{x}$ and $\mathbf{m}$



$$p(r|\mathbf{x} = [3,0]^T, \mathbf{m} = [0,0]^T) = \mathcal{N}(3, 0.3^2) = K \exp\{-\frac{1}{2}\frac{(3-r)^2}{0{,}3^2}\}$$



Mean range = 3 m, σ = 0.3 m

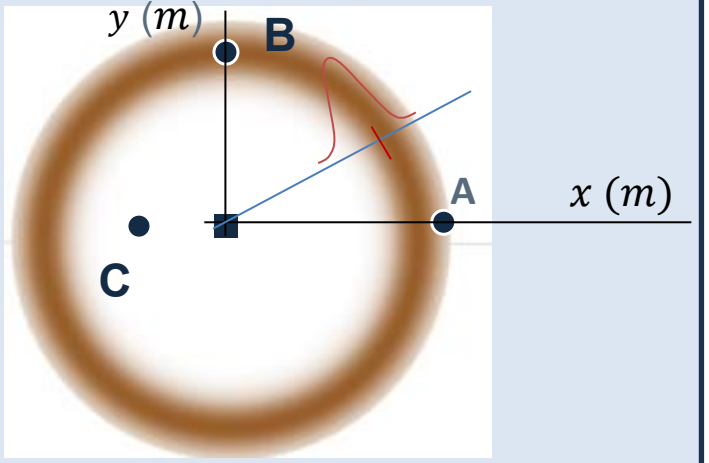
Likelihood space 2D

Compatibility of $\mathbf{x}$ for a given r and $\mathbf{m}$

Data: $r = 3.1\text{m}$ and $\mathbf{m} = [0,0]^T$

$$\mathcal{L}(\boldsymbol{x}; \mathbf{m}, 3.1) = K \cdot \exp\{-\frac{1}{2}\frac{\left([x^2 + y^2]^{1/2} - 3.1\right)^2}{\sigma^2}\}$$

Darker = higher likelihood.
B fits better than **A** an **C**

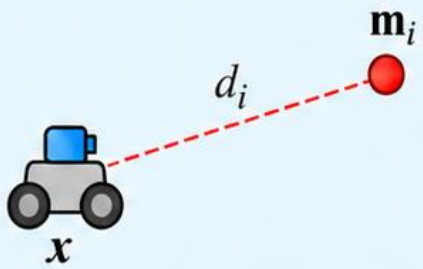


Landmark observation models

- **Map:** collection of landmarks $\mathcal{M} = \{\mathbf{m}_i\}_{i=1,\ldots,N}$.
- A sensor provides measurements to landmarks.

Range

distance to a landmark

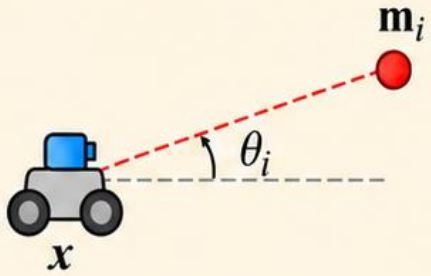


$$z_i = d_i = h_i(\mathbf{x}, \mathbf{m}) + w_i$$

Examples: WiFi, GPS, beacon ranging

Bearing

direction to a landmark



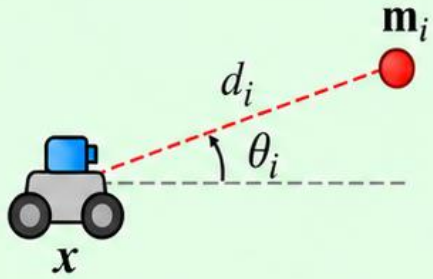
$$z_i = \theta_i = h_i(\mathbf{x}, \mathbf{m}) + w_i$$

(planar world)

Example: monocular camera

Range + bearing

distance and direction



$$\mathbf{z}_i = [d_i,\ \theta_i]^T = \underbrace{\mathbf{h}_i(\mathbf{x}, \mathbf{m}) + \mathbf{w}_i}_{\text{2D vectors}}$$

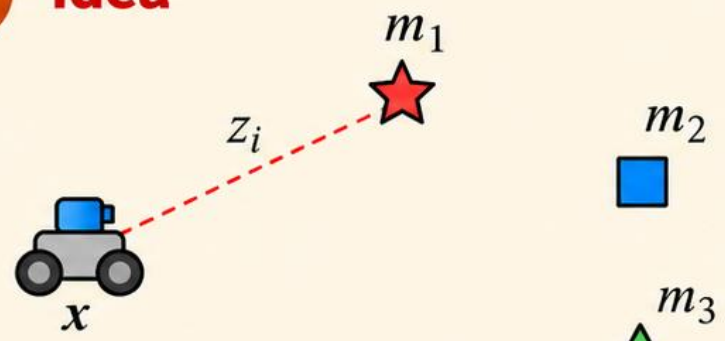
Examples: stereo vision, point-cloud features

Each landmark observation model predicts the expected measurement from robot pose $\mathbf{x}$ and landmark $\mathbf{m}$, plus sensor noise $\mathbf{w}_i$.

The Data Association problem

- **Problem:** decide which landmark $\boldsymbol{m}_j$ generated an observation $\boldsymbol{z}_i$.
- We need the **correspondence** $z_i \leftrightarrow m_j$ before applying the observation model.

1 Idea

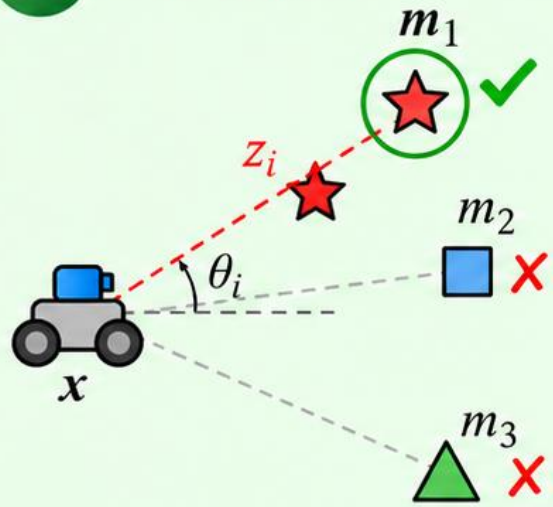


$$z_i = \boldsymbol{h}(\mathbf{x}, \mathbf{m}_j) + w_i$$

Which landmark is it?

data association: $z_i \rightarrow m_j$

2 Example



1 Descriptor: compare appearance / ID

e.g. observed z_i — map landmark $\boldsymbol{m}_j$



2 Geometry: compare predicted and observed range-bearing

e.g. $z_i = [d_i,\ \theta_i]^T$ vs. $\hat{z}_j = \boldsymbol{h}(\mathbf{x}, \mathbf{m}_j)$

Choose the landmark most compatible with the observation.

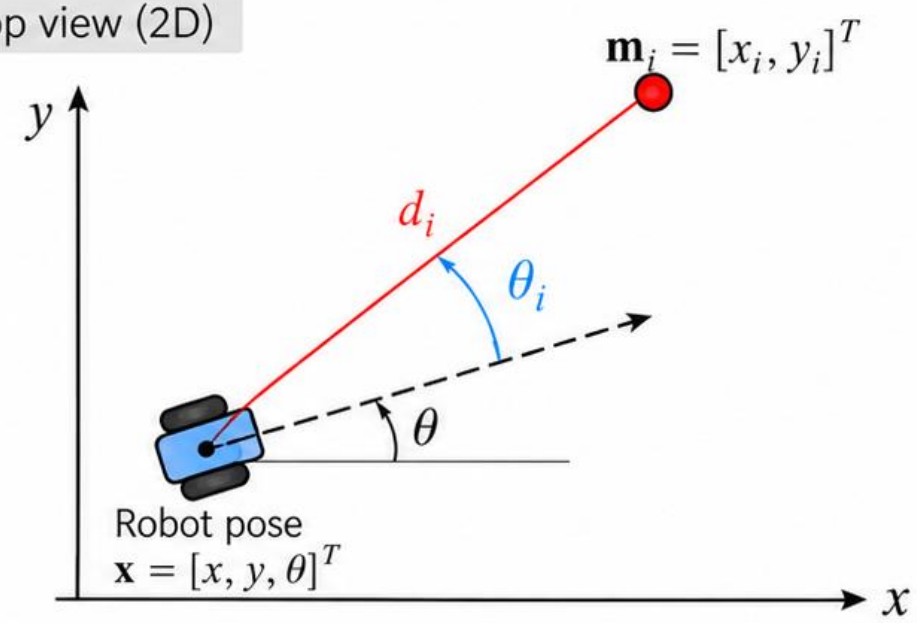
Wrong associations can lead to wrong localization or mapping.
Typical aids: landmark descriptors + geometric compatibility tests.

Landmark observation models: **Distance and bearing** (in 2D)

The observation to a landmark $\mathbf{m}_i$ consists of a range (distance) d_i and a bearing (angle) θ_i :

$$\mathbf{z}_i = \begin{bmatrix} d_i \\ \theta_i \end{bmatrix} = \mathbf{h}(\mathbf{x}, \mathbf{m}_i) + \mathbf{w}_i$$

Top view (2D)



Measurement function

Predicted range and bearing to $\mathbf{m}_i$ from pose $\mathbf{x} = [x, y, \theta]^T$:

$$\mathbf{h}(\mathbf{x}, \mathbf{m}_i) = \begin{bmatrix} \hat{d}_i \\ \hat{\theta}_i \end{bmatrix} = \begin{bmatrix} \sqrt{(x_i - x)^2 + (y_i - y)^2} \\ \text{atan2}(y_i - y, x_i - x) - \theta \end{bmatrix}$$

Observation model

Measurement with additive Gaussian noise:

$$\mathbf{z}_i = \begin{bmatrix} d_i \\ \theta_i \end{bmatrix} = \mathbf{h}(\mathbf{x}, \mathbf{m}_i) + \mathbf{w}_i \qquad \mathbf{w}_i \sim \mathcal{N}(0, \mathbf{Q}_i)$$

Q_i : 2×2 covariance matrix (range and bearing noise)

Likelihood

Probability of measuring $\mathbf{z}_i$ given $\mathbf{x}$ and $\mathbf{m}_i$:

$$p(\mathbf{z}_i \,|\, \mathbf{x}, \mathbf{m}_i) = \frac{1}{2\pi |\mathbf{Q}_i|^{1/2}} \exp\left\{ -\frac{1}{2} \left[\mathbf{z}_i - \mathbf{h}(\mathbf{x}, \mathbf{m}_i) \right]^T Q_i^{-1} \left[\mathbf{z}_i - \mathrm{h}(\mathbf{x}, \mathbf{m}_i) \right] \right\}$$

Notation

$\mathbf{x} = [x, y, \theta]^T$	robot pose (in 2D)
$\mathbf{m}_i = [x_i, y_i]^T$	i-th landmark (known in the map)
d_i	range (distance) to landmark $\mathbf{m}_i$
θ_i	bearing (measured from robot heading θ)
$\mathbf{z}_i = \begin{bmatrix} d_i \\ \theta_i \end{bmatrix}$	observation vector (2D)
$\mathbf{w}_i \sim \mathcal{N}(0, \mathbf{Q}_i)$	measurement noise

Distance and bearing. Example

1. Estimating the position of an observed landmark

Problem A robot at pose $\mathbf{x} = [x, y, \theta]^T$ observes a landmark $\mathbf{m}_i$ with a range-and-bearing sensor.

Robot pose (global frame)

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} 2\,\text{m} \\ 1\,\text{m} \\ 45^\circ \end{bmatrix}$$

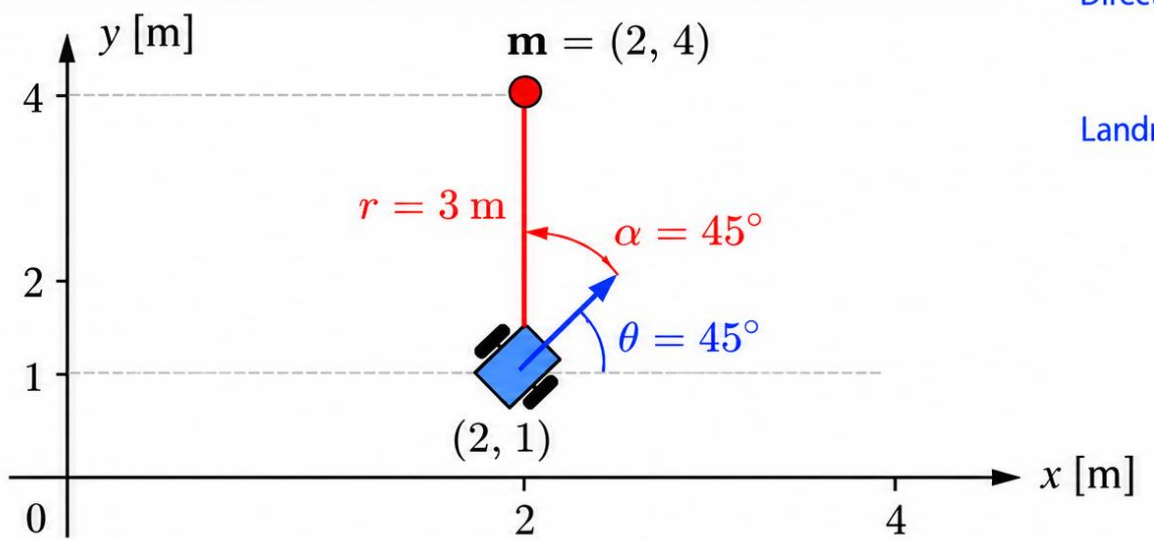
Range–bearing observation

$$\mathbf{z} = \begin{bmatrix} r \\ \alpha \end{bmatrix} = \begin{bmatrix} 3\,\text{m} \\ 45^\circ \end{bmatrix}$$

α : bearing measured from the robot heading θ

Question 1: What is the position of the observed landmark $\mathbf{m}_i$ in the global reference frame?

Solution



Direction to the landmark in the global frame:

$$\phi = \theta + \alpha = 45^\circ + 45^\circ = 90^\circ$$

Landmark position:

$$\mathbf{m} = f(\mathbf{x}, z) = \begin{bmatrix} x + r\cos(\theta + \alpha) \\ y + r\sin(\theta + \alpha) \end{bmatrix}$$

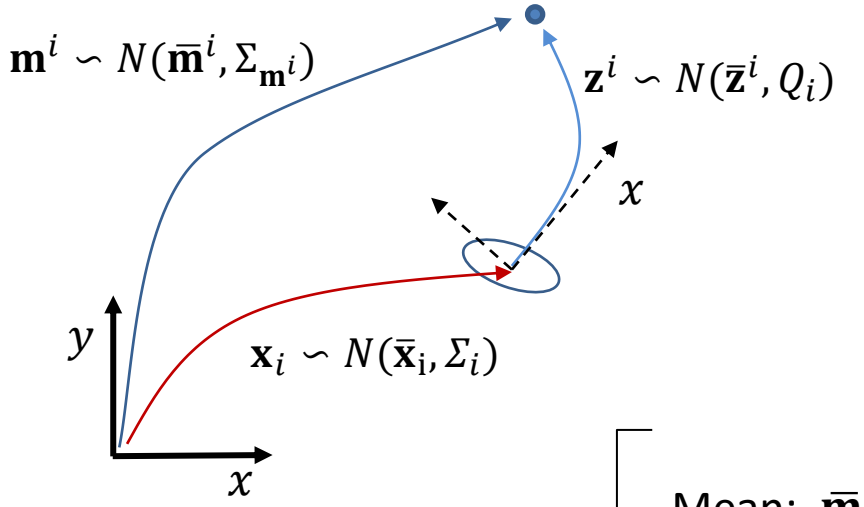
$$\mathbf{m} = \begin{bmatrix} 2 + 3\cos 90^\circ \\ 1 + 3\sin 90^\circ \end{bmatrix} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \text{m}$$

Estimated landmark position:

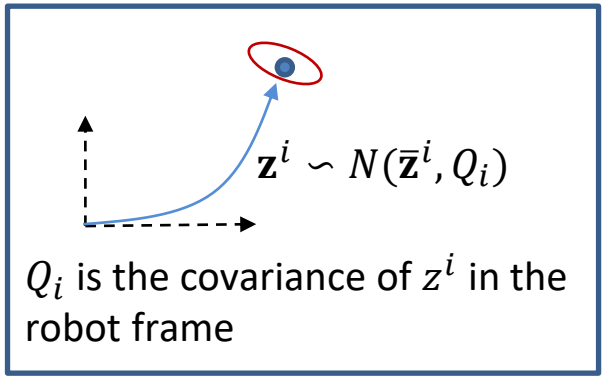
$$\mathbf{m} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \text{m}$$

From measurement uncertainty to **landmark-position uncertainty**

COVARIANCE of an observed 2D landmark (in **cartesian** coordinates)



Observation



For coherence with later notation we have changed from subscript ($\mathbf{z}_i$) to superscript $\mathbf{z}^i$

$$\mathbf{m}^i = \mathbf{x}_i \oplus \mathbf{z}^i = \mathbf{f}(\mathbf{x}_i, \mathbf{z}^i)$$

Mean: $\bar{\mathbf{m}}^i = \bar{\mathbf{x}}_i \oplus \bar{\mathbf{z}}^i$

Covariance: $\underbrace{\Sigma_{\mathbf{m}^i}}_{2\text{x}2} = \underbrace{\frac{\partial \mathbf{m}^i}{\partial \mathbf{x}_i}}_{2\text{x}3} \Sigma_i \left(\frac{\partial \mathbf{m}^i}{\partial \mathbf{x}_i}\right)^T + \underbrace{\frac{\partial \mathbf{m}^i}{\partial \mathbf{z}^i}}_{2\text{x}2} Q_i \left(\frac{\partial \mathbf{m}^i}{\partial \mathbf{z}^i}\right)^T$

uncertainty in m^i due to uncertainty in $\mathbf{x}_i$

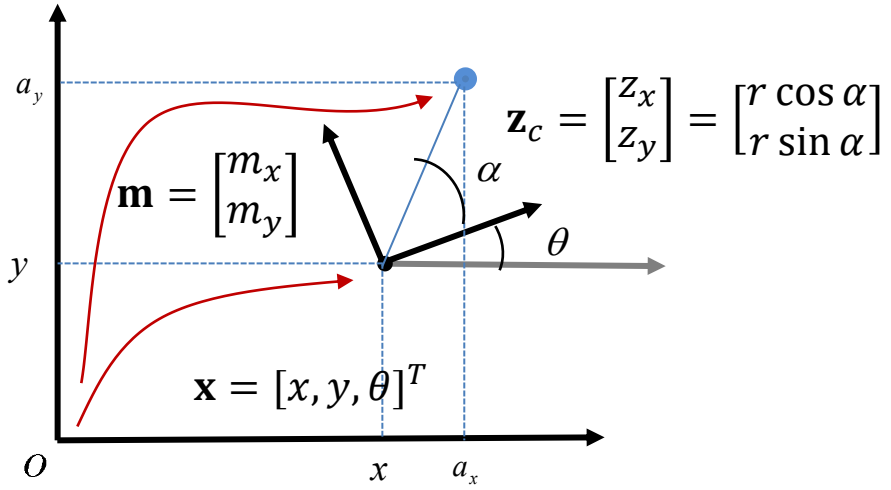
uncertainty in m^i due to uncertainty in $\boldsymbol{z}^i$

RECALL: if $Z=f(X,Y)$

$$\Sigma_Z = \frac{\partial f}{\partial X} \Sigma_X \left(\frac{\partial f}{\partial X}\right)^T + \frac{\partial f}{\partial Y} \Sigma_Y \left(\frac{\partial f}{\partial Y}\right)^T$$

$\Sigma_{\mathbf{m}^i}$ is the sum of two covariance matrices!

COVARIANCE of an observed 2D landmark (in **polar** coordinates)



Sensor pose: $\mathbf{x} = [x, y, \theta]^T$

Cartesian observation: $\mathbf{z_c} = [z_x \quad z_y]^T$

Polar observation: $\mathbf{z_p} = [r \quad \alpha]^T$

$$\mathbf{m} = \mathbf{x} \oplus \mathbf{z_c} = \mathbf{f}(\mathbf{x}, \mathbf{z_c}) = \begin{bmatrix} x + z_x\cos\theta - z_y\sin\theta \\ y + z_x\sin\theta + z_y\cos\theta \end{bmatrix}$$

$$\Sigma_{\mathbf{m}} = \frac{\partial \mathbf{m}}{\partial \mathbf{x}} \Sigma_{\mathbf{x}} \left(\frac{\partial \mathbf{m}}{\partial \mathbf{x}}\right)^{\mathrm{T}} + \frac{\partial \mathbf{m}}{\partial \mathbf{z_c}} \Sigma_{\mathbf{z_c}} \left(\frac{\partial \mathbf{m}}{\partial \mathbf{z_c}}\right)^{\mathrm{T}}$$

Since

$$\mathbf{z_c} = \begin{bmatrix} z_x \\ z_y \end{bmatrix} = \begin{bmatrix} r\cos\alpha \\ r\sin\alpha \end{bmatrix} = \mathbf{f}(\mathbf{z_p}) \quad \Rightarrow \quad \Sigma_{\mathbf{z_c}} = \frac{\partial \mathbf{z_c}}{\partial \mathbf{z_p}} \Sigma_{\mathbf{z_p}} \left(\frac{\partial \mathbf{z_c}}{\partial \mathbf{z_p}}\right)^{\mathrm{T}}$$

with $\frac{\partial \mathbf{z_c}}{\partial \mathbf{z_p}} = \begin{bmatrix} cos\alpha & -rsin\alpha \\ sin\alpha & rcos\alpha \end{bmatrix}$

Chain rule of a composite function

$$\frac{\partial \mathbf{m}}{\partial \mathbf{z_c}} \frac{\partial \mathbf{z_c}}{\partial \mathbf{z_p}} \Sigma_{\mathbf{z_p}} \left(\frac{\partial \mathbf{z_c}}{\partial \mathbf{z_p}}\right)^{\mathbf{T}} \left(\frac{\partial \mathbf{m}}{\partial \mathbf{z_c}}\right)^{\mathbf{T}} = \frac{\partial \mathbf{m}}{\partial \mathbf{z_p}} \Sigma_{\mathbf{z_p}} \left(\frac{\partial \mathbf{m}}{\partial \mathbf{z_p}}\right)^{\mathrm{T}}$$

We can use either of two expressions

Distance and bearing. Example

2. Uncertainty of the estimated landmark position

Problem The robot pose and the range–bearing measurement are uncertain. We model them as independent Gaussian random variables.

$$\mathbf{x} \sim \mathcal{N}(\bar{\mathbf{x}}, \Sigma_x) \qquad \mathbf{z} \sim \mathcal{N}(\bar{\mathbf{z}}, \Sigma_z)$$

Uncertainty parameters (standard deviations)

σ_x	$= 0.10$ m	(robot x position)
σ_y	$= 0.10$ m	(robot y position)
σ_θ	$= 5° = 0.0873$ rad	(robot heading)
σ_r	$= 0.20$ m	(measured range)
σ_α	$= 3° = 0.0524$ rad	(measured bearing)

Σ_x (3×3)

$$\Sigma_x = \begin{bmatrix} 0.01 & 0 & 0 \\ 0 & 0.01 & 0 \\ 0 & 0 & 0.0076 \end{bmatrix}$$

Σ_z (2×2)

$$\Sigma_z = \begin{bmatrix} 0.04 & 0 \\ 0 & 0.0027 \end{bmatrix}$$

Uncertainty propagation

Landmark position:

$$\mathbf{m}_i = f(\mathbf{x}, \mathbf{z}) = \begin{bmatrix} x + r\cos(\theta + \alpha) \\ y + r\sin(\theta + \alpha) \end{bmatrix}$$

First-order approximation:

$$\Sigma_m \approx \mathbf{J}_x \Sigma_x \mathbf{J}_x^T + \mathbf{J}_z \Sigma_z \mathbf{J}_z^T$$

Jacobian w.r.t. robot pose $\mathbf{x} = [x, y, \theta]^T$

$$\mathbf{J}_x = \frac{\partial f}{\partial \mathbf{x}} = \begin{bmatrix} 1 & 0 & -r\sin(\theta + \alpha) \\ 0 & 1 & r\cos(\theta + \alpha) \end{bmatrix}$$

For $r = 3$ m, $\theta + \alpha = 90°$:

$$\mathbf{J}_x = \begin{bmatrix} 1 & 0 & -3 \\ 0 & 1 & 0 \end{bmatrix}$$

Jacobian w.r.t. measurement $\mathbf{z} = [r, \alpha]^T$

$$\mathbf{J}_z = \frac{\partial f}{\partial \mathbf{z}} = \begin{bmatrix} \cos(\theta + \alpha) & -r\sin(\theta + \alpha) \\ \sin(\theta + \alpha) & r\cos(\theta + \alpha) \end{bmatrix}$$

For $r = 3$ m, $\theta + \alpha = 90°$:

$$\mathbf{J}_z = \begin{bmatrix} 0 & -3 \\ 1 & 0 \end{bmatrix}$$

Result

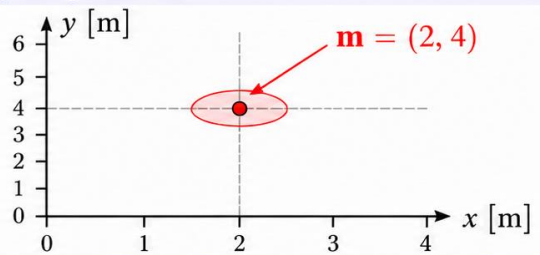
Estimated landmark position:

$$\mathbf{m} = \begin{bmatrix} 2 \\ 4 \end{bmatrix} \text{m}$$

Covariance of the estimated position:

$$\Sigma_{\mathbf{m}} \approx \begin{bmatrix} 0.1032 & 0 \\ 0 & 0.0500 \end{bmatrix} \text{m}^2$$

Uncertainty ellipse (2σ)



Distance and bearing. Example II

1. Estimating the position of an observed landmark

Problem A robot at pose $\mathbf{x} = [x, y, \theta]^T$ observes a landmark $\mathbf{m}$ with a range-and-bearing sensor.

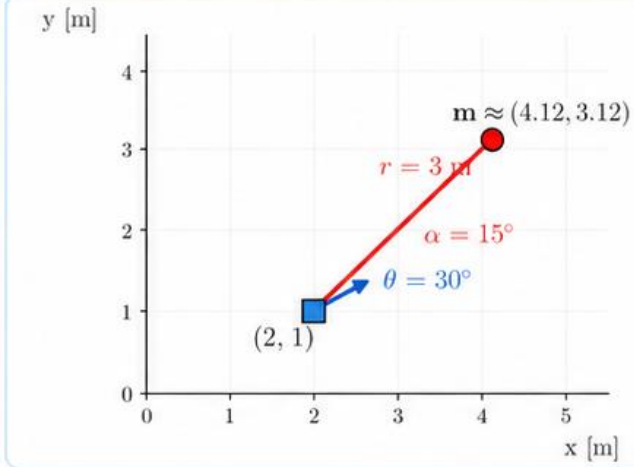
Robot pose (global frame)

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \theta \end{bmatrix} = \begin{bmatrix} 2\text{ m} \\ 1\text{ m} \\ 30^\circ \end{bmatrix}$$

Range–bearing observation

$$\mathbf{z} = \begin{bmatrix} r \\ \alpha \end{bmatrix} = \begin{bmatrix} 3\text{ m} \\ 15^\circ \end{bmatrix}$$

α: bearing relative to robot heading θ



Question Where is the observed landmark in the global reference frame?

Solution

Direction to landmark (global frame):

$$\phi = \theta + \alpha = 30^\circ + 15^\circ = 45^\circ$$

Landmark position:

$$\mathbf{m} = f(\mathbf{x}, \mathbf{z}) = \begin{bmatrix} x + r\cos(\theta + \alpha) \\ y + r\sin(\theta + \alpha) \end{bmatrix} = \begin{bmatrix} 2 + 3\cos 45^\circ \\ 1 + 3\sin 45^\circ \end{bmatrix}$$

Estimated landmark position

$$\mathbf{m} \approx [4.1213,\ 3.1213]^T \text{ m}$$

2. Uncertainty of the estimated landmark position

Problem Robot pose and range–bearing measurement have independent Gaussian uncertainties. $\mathbf{x} \sim \mathcal{N}(\bar{\mathbf{x}}, \Sigma_x)$ $\mathbf{z} \sim \mathcal{N}(\bar{\mathbf{z}}, \Sigma_z)$

Uncertainty parameters (standard deviations)

- $\sigma_x = 0.10$ m (robot x)
- $\sigma_y = 0.10$ m (robot y)
- $\sigma_\theta = 5^\circ = 0.08727$ rad
- $\sigma_r = 0.20$ m (range)
- $\sigma_\alpha = 3^\circ = 0.05236$ rad

Σ_x (3x3)

$$\begin{bmatrix} 0.01 & 0 & 0 \\ 0 & 0.01 & 0 \\ 0 & 0 & 0.007615 \end{bmatrix}$$

Σ_z (2x2)

$$\begin{bmatrix} 0.0400 & 0 \\ 0 & 0.002742 \end{bmatrix}$$

Uncertainty propagation

Landmark position:

$$\mathbf{m} = f(\mathbf{x}, \mathbf{z}) = \begin{bmatrix} x + r\cos(\theta + \alpha) \\ y + r\sin(\theta + \alpha) \end{bmatrix}$$

First-order approximation:

$$\Sigma_m \approx \mathbf{J}_x \Sigma_x \mathbf{J}_x^T + \mathbf{J}_z \Sigma_z \mathbf{J}_z^T$$

Jacobian w.r.t. robot pose

$$\mathbf{J}_x = \begin{bmatrix} 1 & 0 & -r\sin\phi \\ 0 & 1 & r\cos\phi \end{bmatrix} = \begin{bmatrix} 1 & 0 & -2.1213 \\ 0 & 1 & 2.1213 \end{bmatrix}$$

Jacobian w.r.t. measurement

$$\mathbf{J}_z = \begin{bmatrix} \cos\phi & -r\sin\phi \\ \sin\phi & r\cos\phi \end{bmatrix} = \begin{bmatrix} 0.7071 & -2.1213 \\ 0.7071 & 2.1213 \end{bmatrix}$$

Result

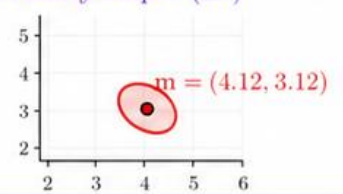
Estimated landmark position:

$$\mathbf{m} = \begin{bmatrix} 4.1213 \\ 3.1213 \end{bmatrix} \text{ m}$$

Covariance:

$$\Sigma_m \approx \begin{bmatrix} 0.0766 & -0.0266 \\ -0.0266 & 0.0766 \end{bmatrix} \text{ m}^2$$

Uncertainty ellipse (2σ)



Probabilistic Sensor Models

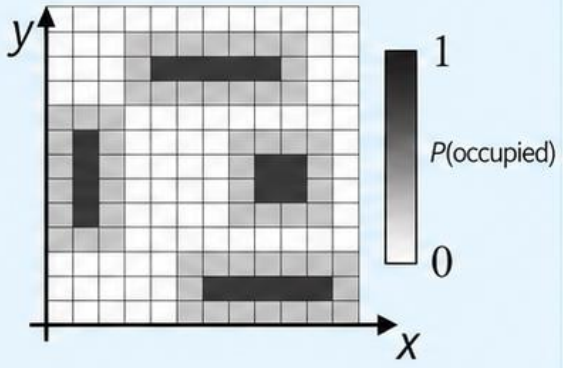
Scan observation models

Map representations

Segment-based map

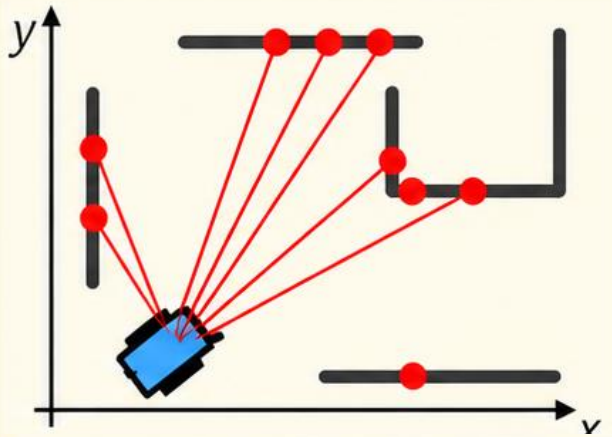


Grid map



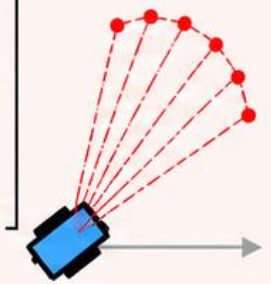
Sensor model $p(\mathbf{z}|\mathbf{x},\mathbf{m})$

How likely is a **scan z**, given a robot pose $\mathbf{x}$ and map $\mathbf{m}$?



Laser scan z

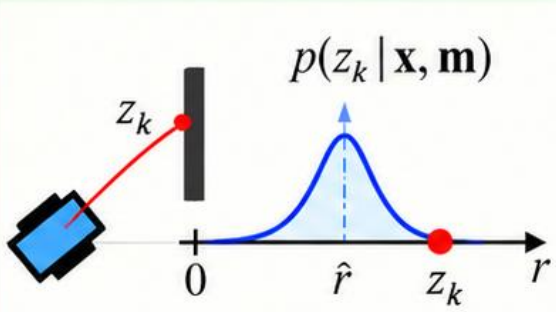
$$\mathbf{z} = \begin{bmatrix} z_1 \\ z_2 \\ \vdots \\ z_K \end{bmatrix}$$



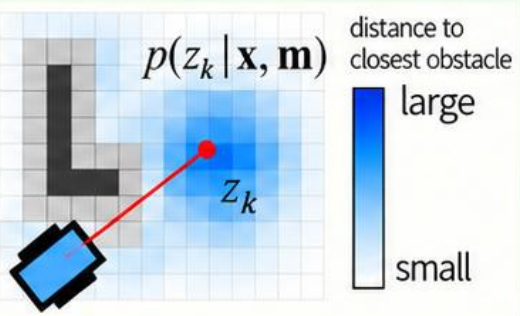
range scan (K beams)

Two common observation models for scans

Beam model

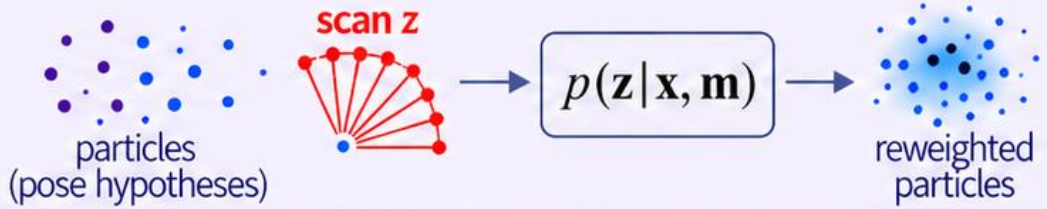


Likelihood field model



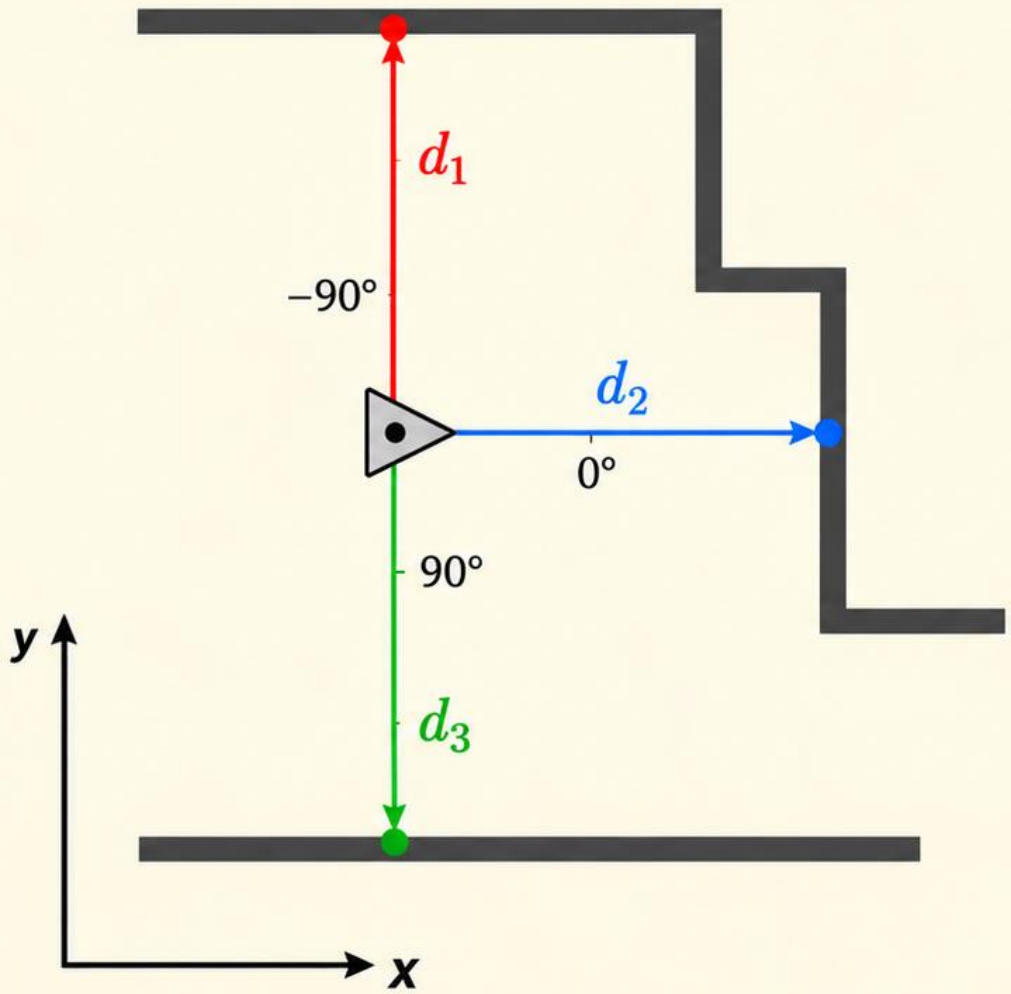
Applications

These observation models are mostly used for localization with particle filters. For example, the localization method **AMCL in ROS 2** supports both types of sensor models.



Beam model

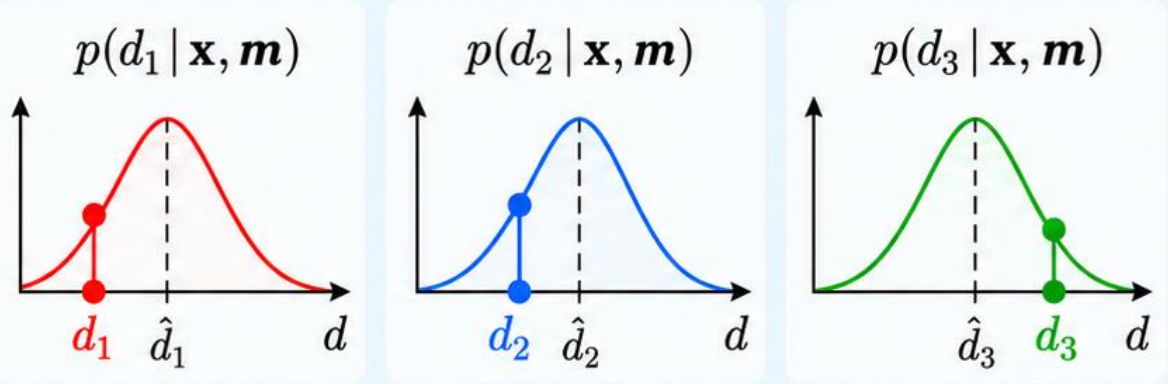
Example: 3 range sensors (laser scan)



Likelihood of each observed beam

For a candidate pose $\mathbf{x}$ and map $\boldsymbol{m}$, each beam has a predicted range $\hat{d}_i = h_i(\mathbf{x}, \boldsymbol{m})$.

The likelihood $p(d_i \mid \mathbf{x}, \boldsymbol{m})$ is the value of the Gaussian at the observed measurement d_i.

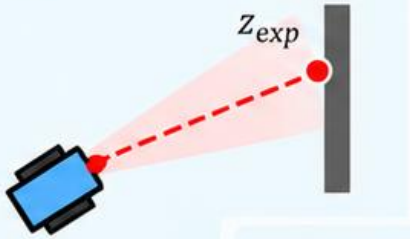


Beam model likelihood

$$p(d_1, d_2, d_3 \mid \mathbf{x}, \boldsymbol{m}) = \prod_{k=1}^{3} p(d_k \mid \mathbf{x}, \boldsymbol{m})$$

Beam model for $p(d_k|\boldsymbol{x}, \boldsymbol{m})$

1 Hit: measurement around expected range

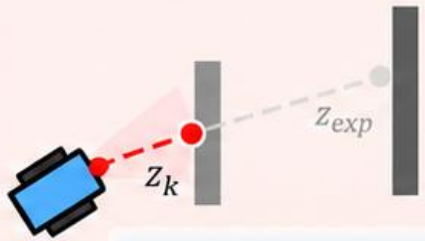


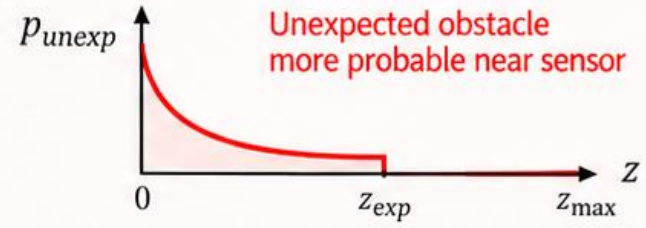


$$p_{hit}(z \mid \mathbf{x}, \mathbf{m}) = \eta_{hit} \frac{1}{\sigma\sqrt{2\pi}} \exp\left(-\frac{(z - z_{exp})^2}{2\sigma^2}\right)$$

η_{hit} normalizes the truncated Gaussian to $[0, z_{max}]$.

2 Unexpected hit: obstacle before expected range

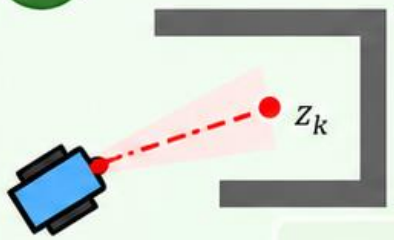


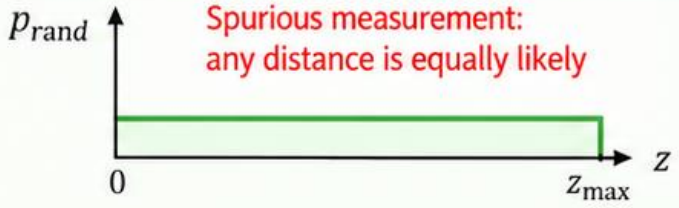


$$p_{unexp}(z \mid \mathbf{x}, \mathbf{m}) = \begin{cases} \eta_{unexp}\, \lambda e^{-\lambda z} & 0 \le z < z_{exp} \\ 0 & otherwise \end{cases}$$

η_{unexp} normalizes the truncated exponential to $[0, z_{exp}]$.

3 Random measurement

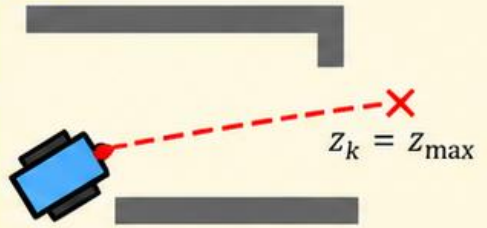


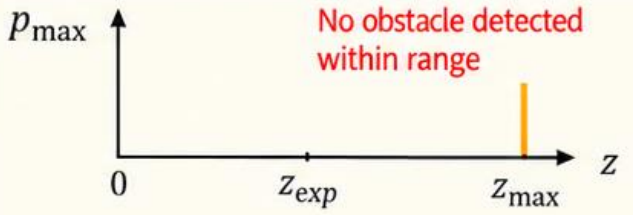


$$p_{rand}(z \mid \mathbf{x}, \mathbf{m}) = \begin{cases} \frac{1}{z_{max}} & 0 \le z < z_{max} \\ 0 & otherwise \end{cases}$$

Models random, spurious readings unrelated to the expected geometry (e.g. interference, multi-path, sensor errors).

4 Max range: no echo received



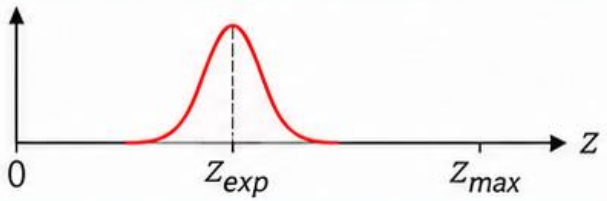


$$p_{max}(z \mid \mathbf{x}, \mathbf{m}) = \begin{cases} 1, & z = z_{max} \\ 0, & otherwise \end{cases}$$

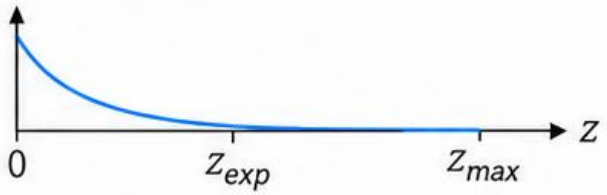
z_{exp} is obtained by ray casting over the known map $\boldsymbol{m}$ (given $\boldsymbol{x}$).

Beam model for $p(d_k|\boldsymbol{x}, \boldsymbol{m})$

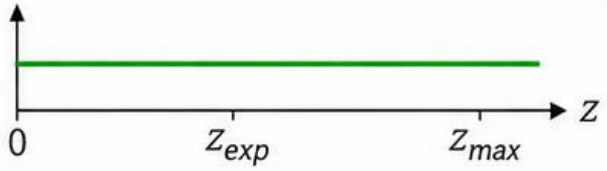
Measurement noise



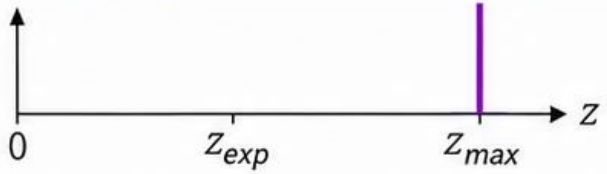
Unexpected obstacles



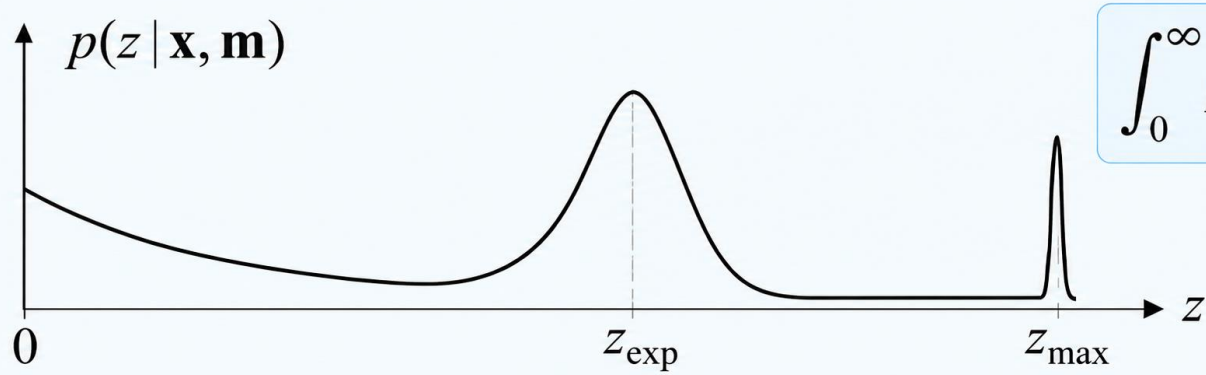
Random measurements



Max range



Combined beam model $p(z \mid \mathbf{x}, \mathbf{m})$



$$\int_0^\infty p(z \mid \mathbf{x}, \mathbf{m})\, dz = 1$$

Mixture model

$$p(z \mid \mathbf{x}, \mathbf{m}) = \alpha_{hit} p_{hit}(z \mid \mathbf{x}, \mathbf{m}) + \alpha_{unexp} p_{unexp}(z \mid \mathbf{x}, \mathbf{m}) + \alpha_{rand} p_{rand}(z \mid \mathbf{x}, \mathbf{m}) + \alpha_{max} p_{max}(z \mid \mathbf{x}, \mathbf{m})$$

Compute $p(z \mid \mathbf{x}, \mathbf{m})$ for a beam

1. Ray cast from pose **x** in map **m** (**ray casting**)
2. Get expected distance z_{exp}
3. Evaluate $p(z \mid \mathbf{x}, \mathbf{m})$ using the mixture model

→ **Costly for laser scanners** (many rays)

Beam model → multimodal, non-Gaussian sensor model

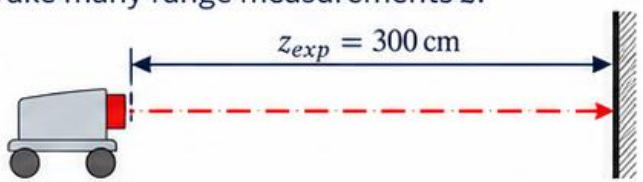
Beam model: learning the mixture parameters

$$p(z_k|\boldsymbol{x},\boldsymbol{m}) = \alpha_{hit}\, p_{hit}(z_k|\boldsymbol{x},\boldsymbol{m}) + \alpha_{unexp}\, p_{unexp}(z_k|\boldsymbol{x},\boldsymbol{m}) + \alpha_{rand}\, p_{rand}(z_k|\boldsymbol{x},\boldsymbol{m}) + \alpha_{max}\, p_{max}(z_k|\boldsymbol{x},\boldsymbol{m}) \quad \Big| \quad \sum_i \alpha_i = 1$$

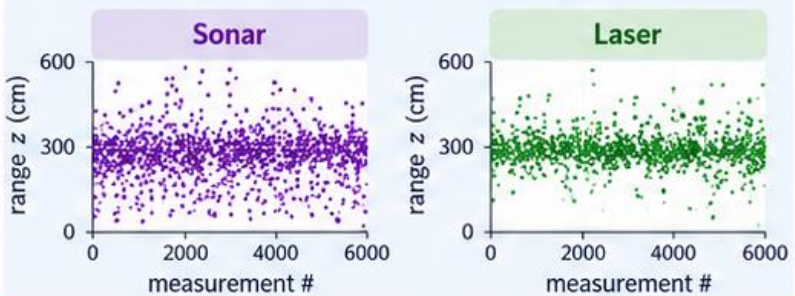
How do we estimate α_{hit}, α_{unexp}, α_{rand} and α_{max} from real data?

1 Collect calibration data

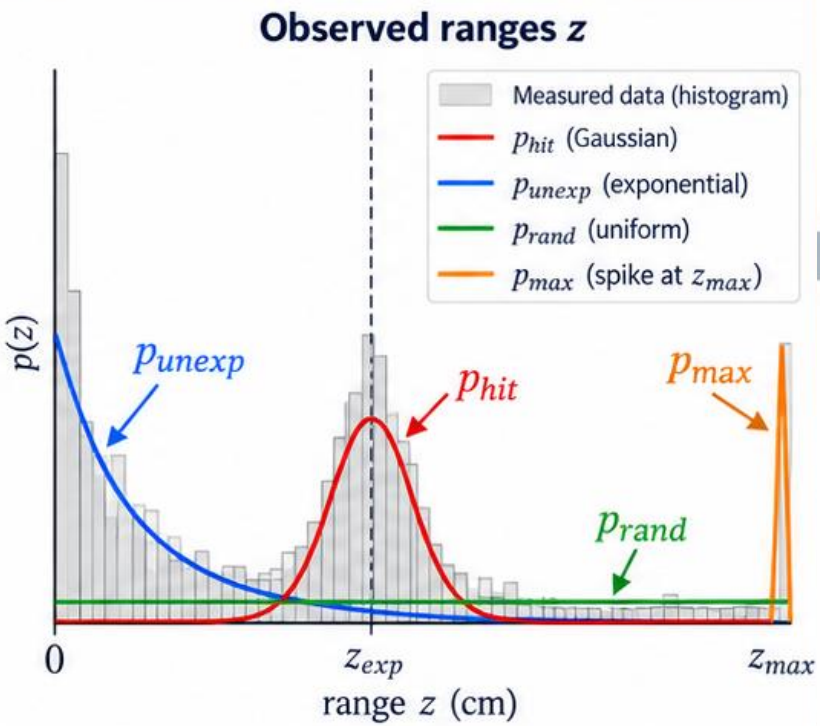
- Place robot / range sensor at a known distance in a known map (e.g., facing a wall).
- Ray-cast in the map to obtain expected range z_{exp} (e.g., $z_{exp} = 300\,\text{cm}$).
- Take many range measurements z.



Repeated measurements (examples)



2 Fit the mixture model



3 Estimate parameters

- Use maximum likelihood to fit the model parameters to the collected data:

$$\Theta^* = \arg\max_{\Theta} \sum_i \log p(z^{(i)}|\boldsymbol{x}^{(i)}, \boldsymbol{m}; \Theta)$$

$$\Theta = \{\alpha_{hit}, \alpha_{unexp}, \alpha_{rand}, \alpha_{max}, \sigma_{hit}, \lambda_{unexp}\}$$

Typical solution: Expectation–Maximization (EM)

Initialize → Set initial values for Θ

E-step → Assign responsibilities to mixture components

M-step → Update α's and shape parameters

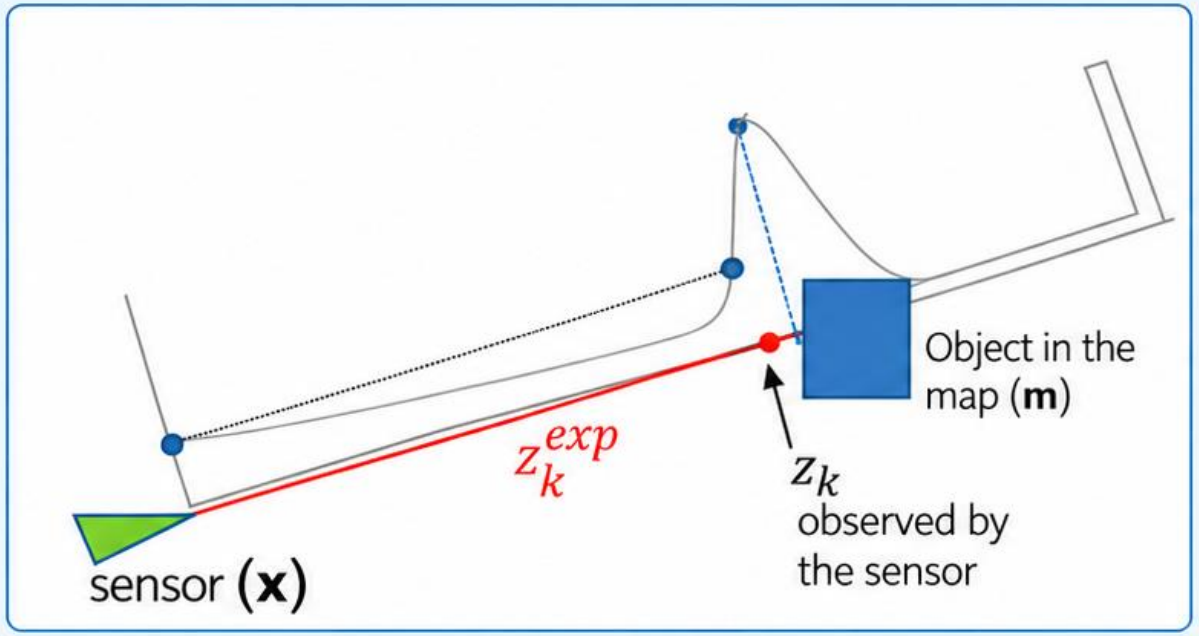
Common presentation in robotics courses: Maximum Likelihood, often solved with EM.

Takeaway: The learned weights explain how often each physical cause occurs. In practice, laser data usually yields a stronger ***hit*** component than sonar.

Beam model for $p(\{d_1, .., d_K\}|x, m)$

1 For each beam k (x, m known):

1. Trace the ray from the sensor pose **x** to the map **m** (ray casting).
2. The **first hit** gives the **expected range** z_k^{exp}.



2 Algorithm beam_range_finder_model(x, m):

$q = 1$
for $k = 1$ to K **do**
  Compute z_k^{exp} using ray casting (first hit)
  Evaluate beam likelihood $p(d_k \mid z_k^{exp}) =$

$$\alpha_{hit}\, p_{hit}(d_k \mid z_k^{exp}) + \alpha_{unexp}\, p_{unexp}(d_k) + \alpha_{rand}\, p_{rand}(d_k) + \alpha_{max}\, p_{max}(d_k)$$

  $q = q \cdot p(d_k \mid z_k^{exp})$
end for
return q

3 Likelihood of the complete scan

$$p(\{d_1, \dots, d_K\}|\,\mathbf{x}, \mathbf{m}) = \prod_{k=1}^{K} p(d_k|\,\mathbf{x}, \mathbf{m})$$

Drawbacks of the beam model:

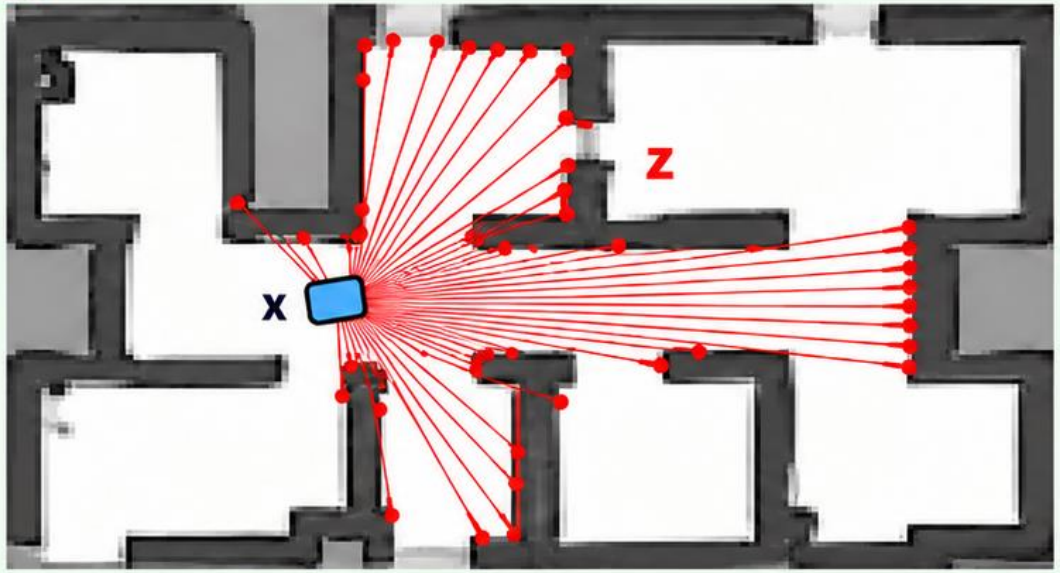
- Not smooth with respect to **x** due to small obstacles and edges in the map.
- Computationally intensive (requires ray casting for each beam).

Beam model for $p(\{d_1, .., d_K\}|x, m)$

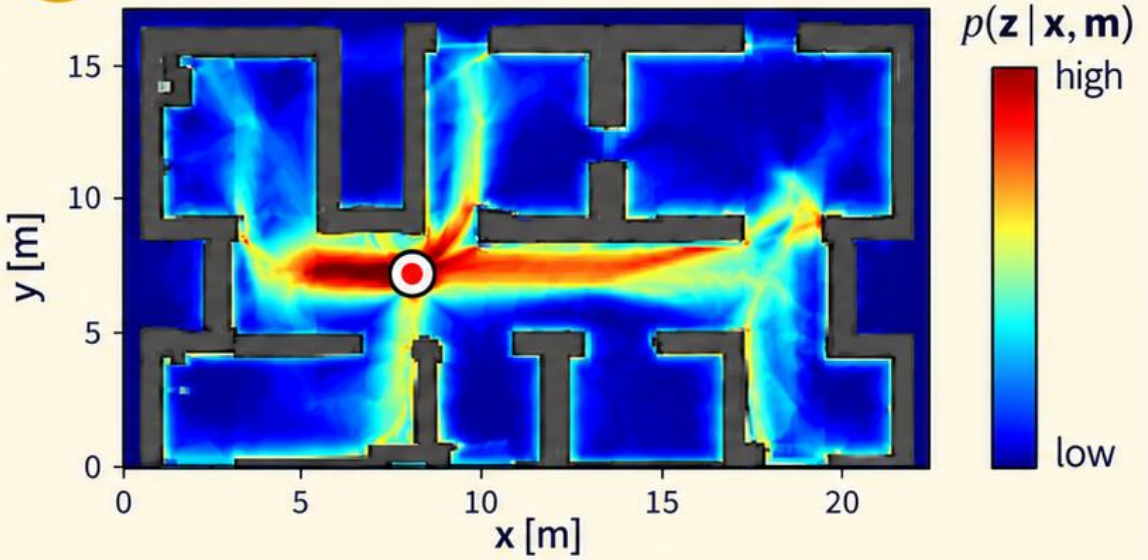
Given a map **m** and a robot pose **x**, the likelihood of a laser scan **z** is the product of the beam model for all beams:

$$p(\mathbf{z} \mid \mathbf{x}, \mathbf{m}) = \prod_{k=1}^{K} p(z_k \mid \mathbf{x}, \mathbf{m})$$

1 Laser scan z



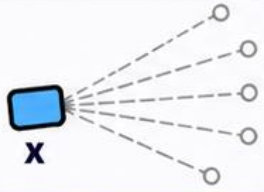
3 Scan likelihood over all poses x



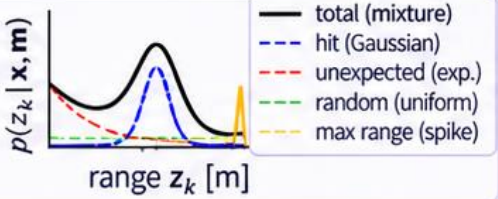
Bright regions correspond to poses whose predicted scan matches the observed scan.

How the likelihood map is computed

1. For each candidate pose **x**, ray-cast the expected scan from the map **m**.



2. Compare each measured beam z_k with its expected beam using $p(z_k \mid \mathbf{x}, \mathbf{m})$.



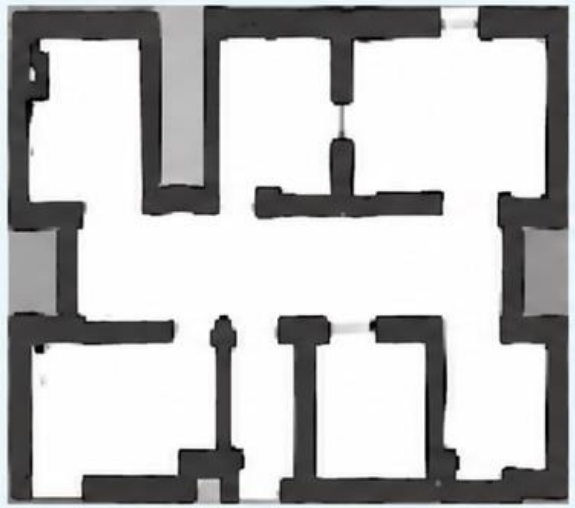
3. Multiply the beam likelihoods to obtain

$$p(\mathbf{z} \mid \mathbf{x}, \mathbf{m}) = \prod_{k=1}^{K} p(z_k \mid \mathbf{x}, \mathbf{m}).$$

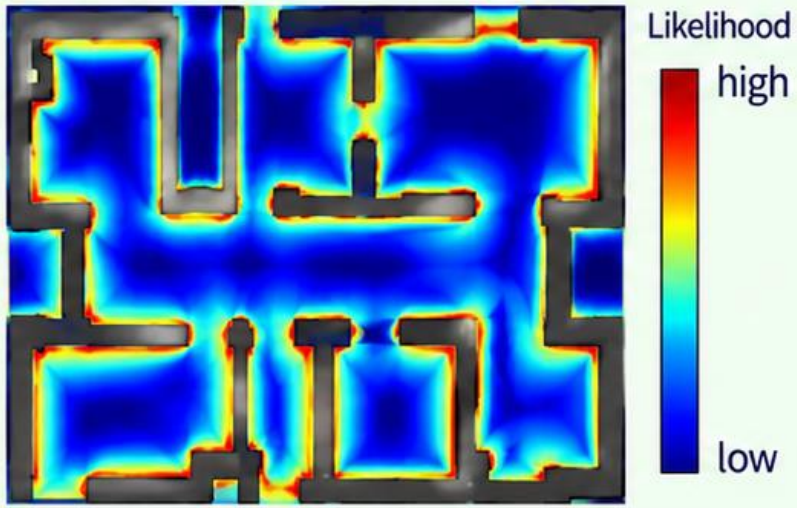
Likelihood Field model

- **Idea:** Instead of following the entire beam, only check the **end points** of the laser beam (this model is also known as **endpoint model**).
- Given a grid map **m**, the likelihood field is computed without any assumption where the sensor is.
- If the map is **static**, the likelihood field is computed only once.

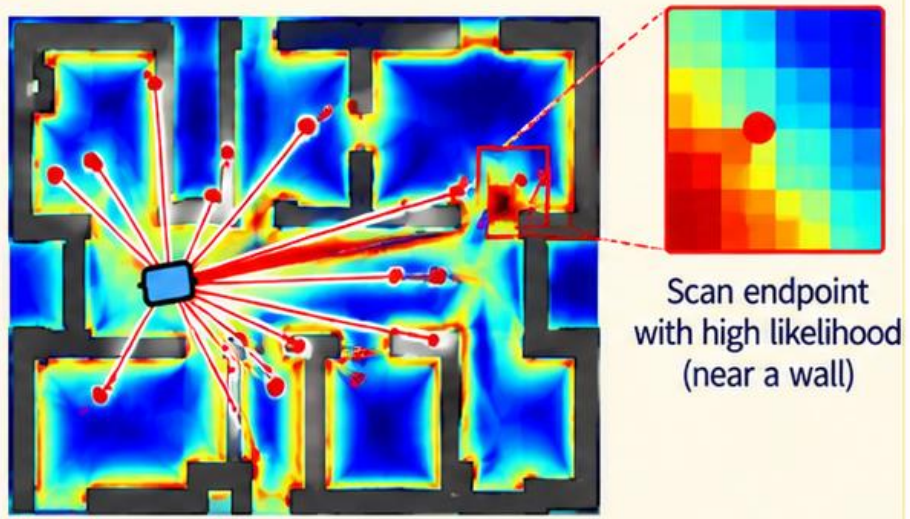
1 **Map m**



2 **Precomputed likelihood field**



3 **Scan endpoints evaluated in LF**

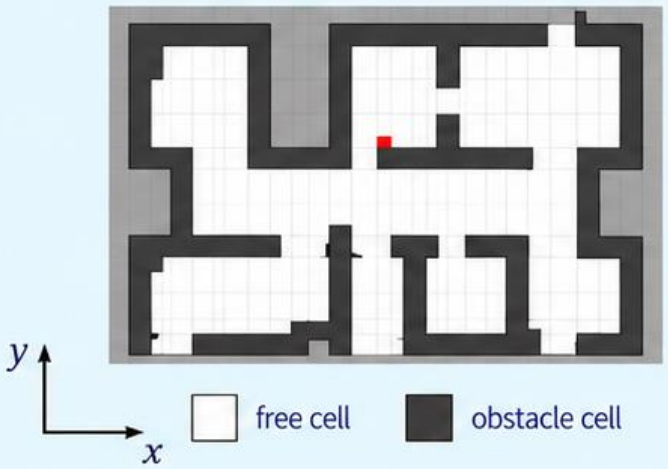


The endpoints of a scan **z** from the map **m** must lie in regions with **high likelihood**.

Likelihood Field model

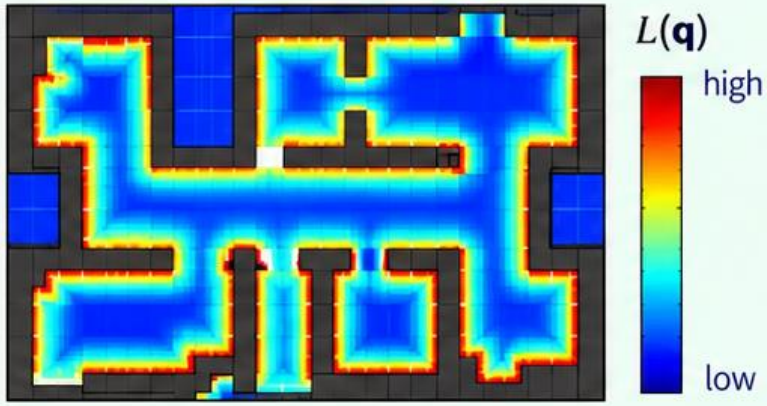
1 Grid map m

Occupancy grid map m (static environment).



2 Precomputed likelihood field $L(\mathbf{q})$

For each cell $\mathbf{q}$ in the grid map, compute the distance to the nearest obstacle and convert it into a likelihood value:



Distance to nearest obstacle:

$$d(\mathbf{q}) = \min_{o \in O} \|\mathbf{q} - \mathbf{o}\|$$

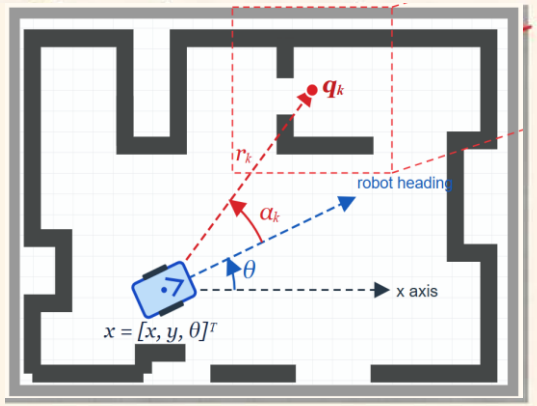
Likelihood field value (Gaussian + uniform):

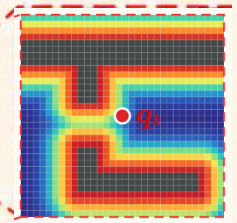
$$L(\mathbf{q}) = \underbrace{\alpha_{hit} \frac{1}{\sqrt{2\pi\sigma}} \exp\left(-\frac{d(\mathbf{q})^2}{2\sigma^2}\right)}_{\text{Gaussian component (obstacle proximity)}} + \underbrace{\alpha_{rand} \frac{1}{z_{max}}}_{\text{Uniform component (random measurements)}}$$

α_{hit} and α_{rand} are model parameters ($\alpha_{hit} + \alpha_{rand} = 1$).

3 Evaluate one beam endpoint

Given a robot pose $x = [x, y, \theta]^T$ and a measurement $\mathbf{z}_k = [r_k, \alpha_k]^T$, compute the endpoint:



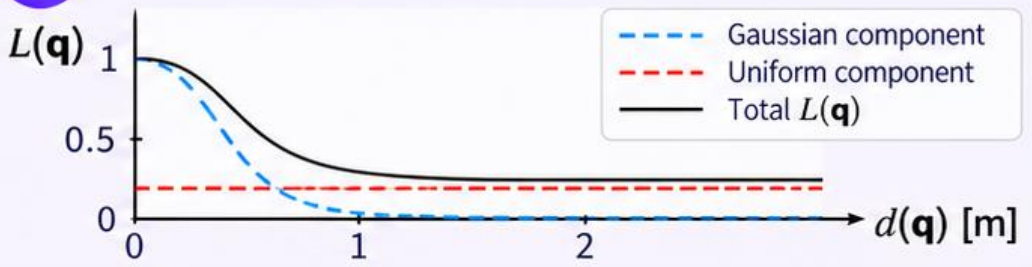


$$\mathbf{q}_k = \begin{bmatrix} x + r_k \cos(\theta + \alpha_k) \\ y + r_k \sin(\theta + \alpha_k) \end{bmatrix}^T$$

$$p(\mathbf{z}_k \mid \mathbf{x}, \mathbf{m}) \approx L(\mathbf{q}_k)$$

The likelihood of this beam is the value of the precomputed field at the endpoint $\mathbf{q}_k$.

4 Likelihood vs. distance to nearest obstacle



5 Full scan likelihood

Assuming conditional independence between beams, the likelihood of a complete scan $\mathbf{z} = \{z_1, \ldots, z_K\}$ is:

$$p(\mathbf{z} \mid \mathbf{x}, \mathbf{m}) \approx \prod_{k=1}^{K} p(z_k \mid \mathbf{x}, \mathbf{m})$$

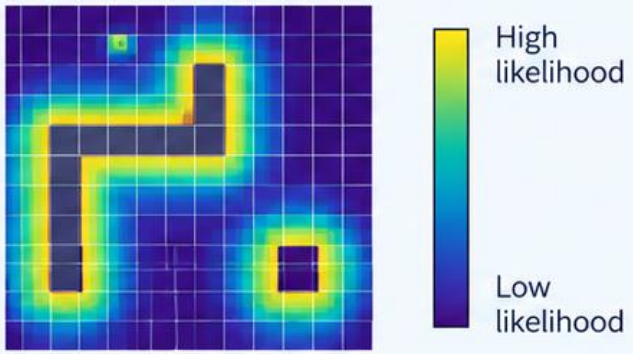
$$\approx \prod_{k=1}^{K} L(\mathbf{q}_k)$$

Likelihood Field model

Properties

1 Highly efficient

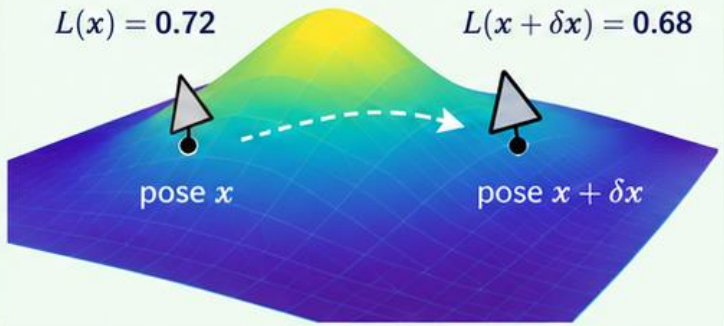
Uses **2D lookup tables** only.



Likelihood field (2D table)

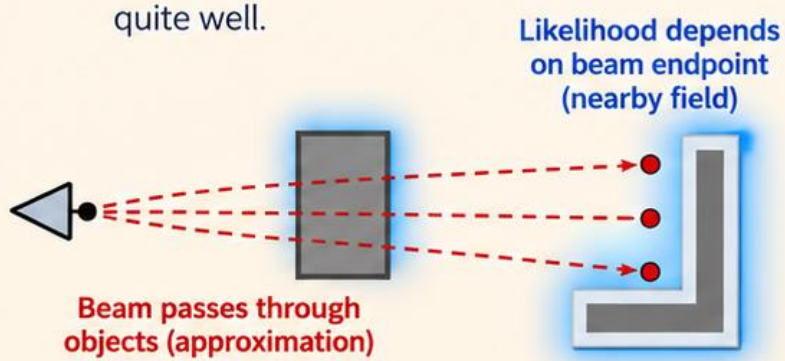
2 Smooth w.r.t. small changes in robot position

Likelihood changes smoothly for small movements.



3 Ignores physical beam properties

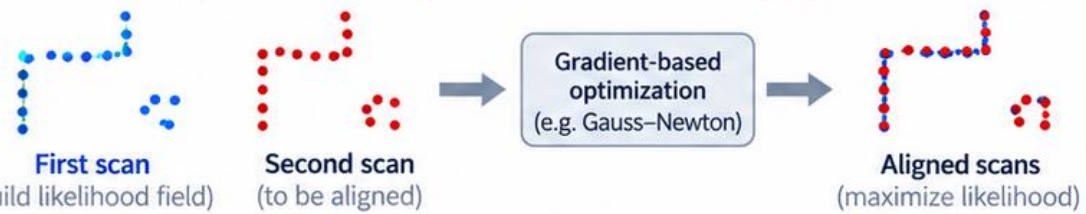
Does not model the full beam path; the sonar/laser beam may pass through objects in the model, but in practice works quite well.



4 Mostly used for

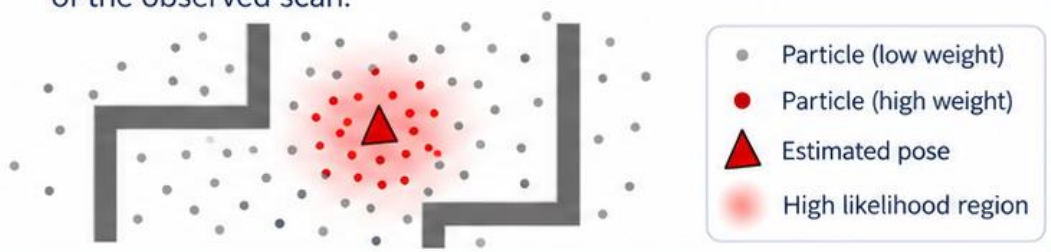
4a Scan matching

One scan defines a likelihood field, and the likelihood of the next scan is optimized with a gradient-based method (e.g. Gauss–Newton).



4b Localization with Particle Filter

Use the likelihood field to weight particles based on the likelihood of the observed scan.



Summary

From physical measurements to probabilistic sensor models

1 Sensors & measurements

- Range, bearing
- GPS, sonar, LiDAR
- Cameras
  - Monocular
  - Stereo
  - RGB-D


LiDAR

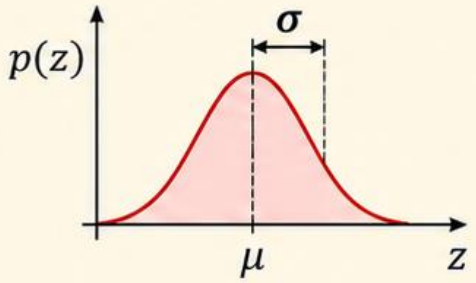

Camera


Sonar


GPS

2 Measurement uncertainty

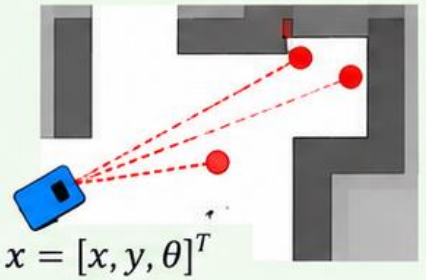
- Repeated measurements are not identical
- Model as a random variable z with realization z
- Typically Gaussian noise



- Multiple measurements and conditional independence

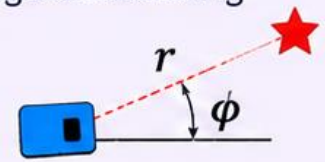
3 Probabilistic sensor model

- Ideal prediction: $z = h(x, m)$
- Sensor model $p(z \mid x, m)$
- PDF over z $\Leftrightarrow$ likelihood over x
- Bayes' theorem explains how the likelihood is used

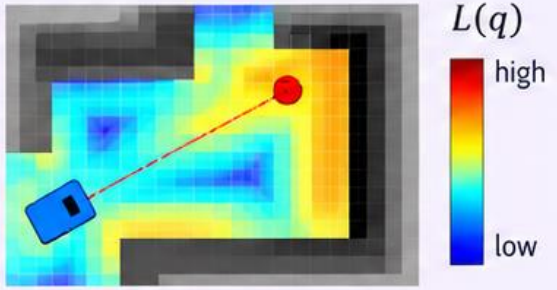


4 Models studied

- Landmark models: range and bearing



- Scan models:
  - Beam model
  - Likelihood-field model



Central idea

$p(z \mid x, m)$ describes how sensor observations depend on the robot pose x and the map m